PERSONAL — BATHROOM DEPLOYMENT

BATHROOM EMERGENCY GUIDE

A Decision-Support Tool for Bathroom Environments

> You're in a bathroom. That's already a good start.

v2.5.0 · 2026-04-29

Bathroom Emergency Guide

🚿 Bathroom Emergency Guide

Version 3.2.0 · 2026-05-03

> "You're in a bathroom. That's already a good start."

A decision-support tool for situations ranging from acute anxiety and physical pain to — hypothetically — zombie apocalypses. Built with scientific rigor and bathroom-tested pragmatism.

Deploy strategically. Read when needed. Breathe always. The math is optional. The breathing is not.

Document Control

Field	Value
Title	Bathroom Emergency Guide
Version	3.2.0
Date	2026-05-03
Classification	Personal — Bathroom Deployment
Build	markdown → Pandoc → HTML/ WeasyPrint PDF/DOCX
Chapters	11 (00–10)
Diagrams	6 original + 8 scientific diagrams
Notation	LaTeX math (\(...\) / \[...\]), Pandoc footnotes
Scientific enhancements	GAD-7, NRS, HRV, cortisol decay, Dunbar numbers, Ostrom's principles, cognitive load theory, Bayes' triage, physiological sigh research, IASC pyramid

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Turn the page to begin.

title: "How to Use This Guide" chapter: 1 revision: "3.2.0" last_updated: "2026-05-03" dependencies: - ../diagrams/master_flowchart.png - ../diagrams/decision_flow_graph.png —

How to Use This Guide

Purpose

This document is a decision-support tool designed for deployment in a bathroom environment. It provides structured guidance for situations ranging from acute anxiety and physical pain to — hypothetically — zombie apocalypses. Every path through this guide converges on one of four practical destinations: the **Calm Guide** (→ Ch.4), the **Self Ambulance Guide** (→ Ch.5), the **Zombie Guide** (→ Ch.6), or the **Professional Support Directory** (→ Ch.7). Regardless of where you enter, you will always be routed toward actionable next steps.

The guide's architecture is formally a directed acyclic graph (DAG) — a structure where information flows in one direction and never loops back on itself (except where explicitly marked 🔁). This means you can enter at any point and still reach a destination without getting stuck in circular reasoning, which is more than most anxious brains can claim.

Target Audience

Anyone currently in a bathroom who needs guidance — emotional, medical, ethical, or existential. No prerequisites beyond physical presence and a willingness to read. The guide assumes no prior training in first aid, psychology, or zombie survival, though it helps if you can operate a door lock.

Mathematical Notation Legend

This guide uses LaTeX math notation to formalize certain concepts — not because bathrooms require calculus, but because precision reduces ambiguity, and ambiguity is the fuel of anxiety. Here's how to read the notation:

Notation	Meaning	Example
`$x$`	Inline math — a variable or formula within text	Heart rate \(HR \in [60, 100]\) bpm
`$$...$$`	Display math — a standalone formula	\[C(t) = C_0 \cdot e^{-\lambda t}\]
`$\in$`	"is an element of" — value belongs to a set	\(NRS \in [0, 10]\)
`$\approx$`	Approximately equal	\(t_{reset} \approx 10\text{min}\)
`$\rightarrow$`	Approaches or leads to	\(n \rightarrow \infty\)

If you see a formula and think “I don’t need this right now” — skip it. The plain-language explanation always follows the math. The formulas are for when you want to understand why the plain language works.

Quick-Start Procedure

1. Identify your current situation from the **Master Flowchart** below (or from the list of seven entry points).
2. Follow the numbered steps along your path. Decisions are marked with ◆; destinations are marked with 📍.
3. Every path ends at one or more destination guides. Go there. Read. Act.
4. If multiple paths apply, handle the most urgent one first (prioritize: physical safety > acute distress > everything else).

Guide Topology — A Graph-Theoretic Model

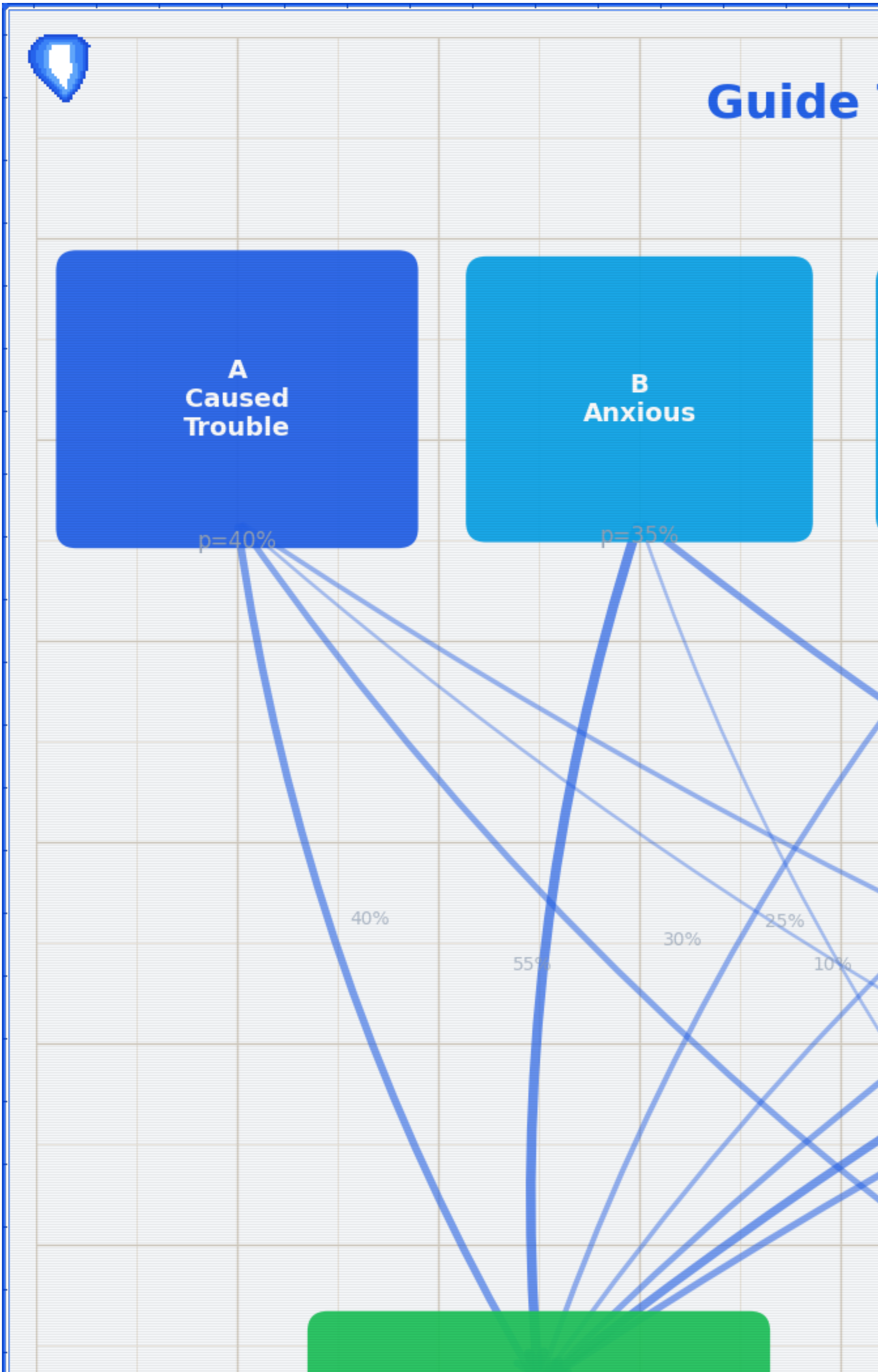
For the mathematically inclined (or the analytically anxious who need to understand the structure before trusting it), this guide can be formalized as a directed graph:

$$G = (V, E)$$

where V is the set of nodes and E is the set of directed edges. Our guide topology has:

- **Entry nodes** ($|V_{entry}| = 7$): Situations A through G — the seven ways you arrived in this bathroom
- **Decision nodes** ($|V_{decision}| \approx 14$): Internal ◆ points within each situation
- **Destination nodes** ($|V_{dest}| = 4$): The four destination guides (Calm, Self Ambulance, Zombie, Professional Support)

The total graph: $|V| = 25$ nodes, $|E| \approx 38$ directed edges. Every path from an entry node reaches at least one destination node — this is the graph’s **reachability invariant**, and it’s what guarantees you’ll never get stuck in this guide without a next step.[1]



Guide
A
Caused
Trouble
p=40%
B
Anxious
p=35%
40%
55%
30%
25%
10%

Decision Flow Graph — structural topology of the guide

Flowchart Legend

Symbol	Meaning
→	Follow this path
◆	Decision point — answer the question to proceed
📍	Destination guide — go to the referenced chapter
§	Terminus — this path has no useful guidance (reconsider your choices)
🔁	Loop back — you may re-enter the flowchart from another entry point

Your Current Status

You are in a bathroom. By definition:

- ✓ The door likely locks — you have privacy
- ✓ Running water is available — you have hygiene resources
- ✓ You are indoors — you are sheltered
- ✓ You chose to come here — some part of you is already practicing self-care

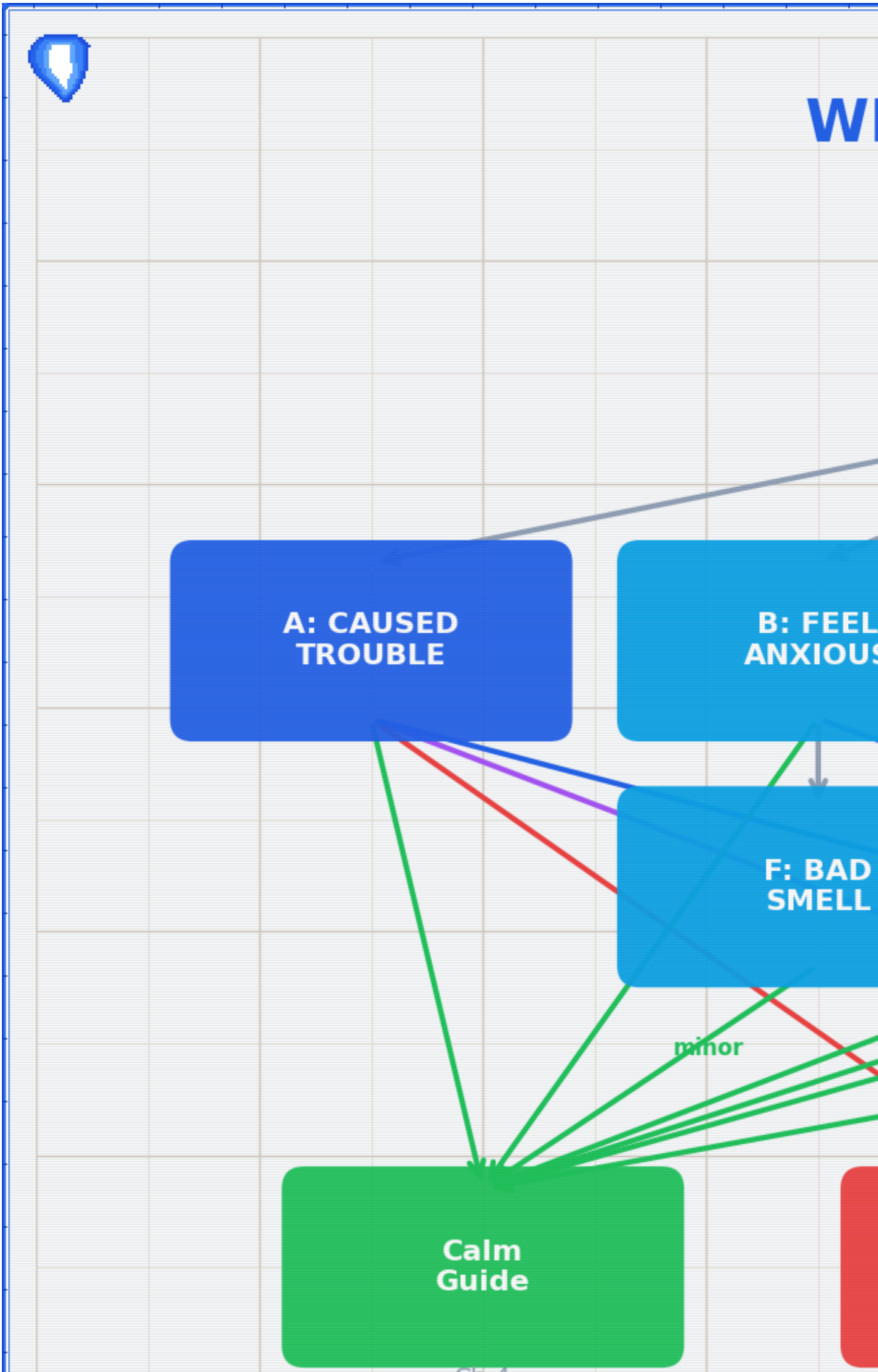
Let's formalize your resource state. Define R as the set of available resources:

$$R = \{privacy, water, shelter, self\_awareness\}$$

With $|R| = 4$, you have more resources than your anxiety is letting you see. Anxiety narrows the perceived resource set to $|R_{perceived}| \ll |R|$. The purpose of this guide is to restore $R_{perceived} \rightarrow R$. This is not metaphor — it's cognitive restructuring, and it works.

This is a stable starting position. Proceed with whatever brought you here.

Master Flowchart — “What Brought You Here?”


A: CAUSED
TROUBLE
B: FEEL
F: BAD
SMELL
minor
Calm
Guide

Master Flowchart — all 7 entry points converging on 4 destination guides

The Seven Entry Points

#	Entry Point	One-line summary	Primary destination
A	Caused Trouble	Something involving another life/entity	Decision tree → varies
B	Feel Anxious	Worry, dread, overthinking	📍 Calm Guide / 📍 Prof. Support
C	Feel Pain	Physical or psychological pain	📍 Self Ambulance / 📍 Calm Guide
D	Feel Endangered	Threat to your safety	📍 Prof. Support / 📍 Calm Guide
E	Things Congesting	Overwhelm, too much at once	📍 Self Ambulance + 📍 Calm Guide
F	Bad Smell	Olfactory emergency	Matches + 📍 Calm Guide (minor)
G	No Place to Go	Isolation, homelessness of spirit	📍 Prof. Support + 📍 Calm Guide

Navigation note. These entry points are not mutually exclusive. If multiple situations apply, start with the one that feels most pressing. The guide is designed so that paths can be combined — for example, if you caused trouble and feel anxious, work through Situation A first (it may route you to the Calm Guide anyway), then address Situation B.

title: “Situation A — I Caused Trouble (Involving a Life/Entity)” chapter: 2 revision: “3.2.0” last_updated: “2026-05-03” dependencies: - ../diagrams/situation_a_tree.png —

Situation A — “I Caused Trouble (Involving a Life/Entity)”

So you’ve affected another sensible form of life — or something that might become one. First: the fact that you’re reading this in a bathroom suggests you haven’t fled the country, which is already a positive sign. Let’s sort out what exactly happened.

Step 1. Identify the nature of the entity.

```
◆ What kind of entity are we talking about?

    ├─ Biological (virus, bacterium, multicellular…)
    │     └─ Does it cause zombification?
    │           ├─ Yes → 🔁 Re-classify as "post-apocalypse" → 📍 Zombie Guide (Ch.6)
```

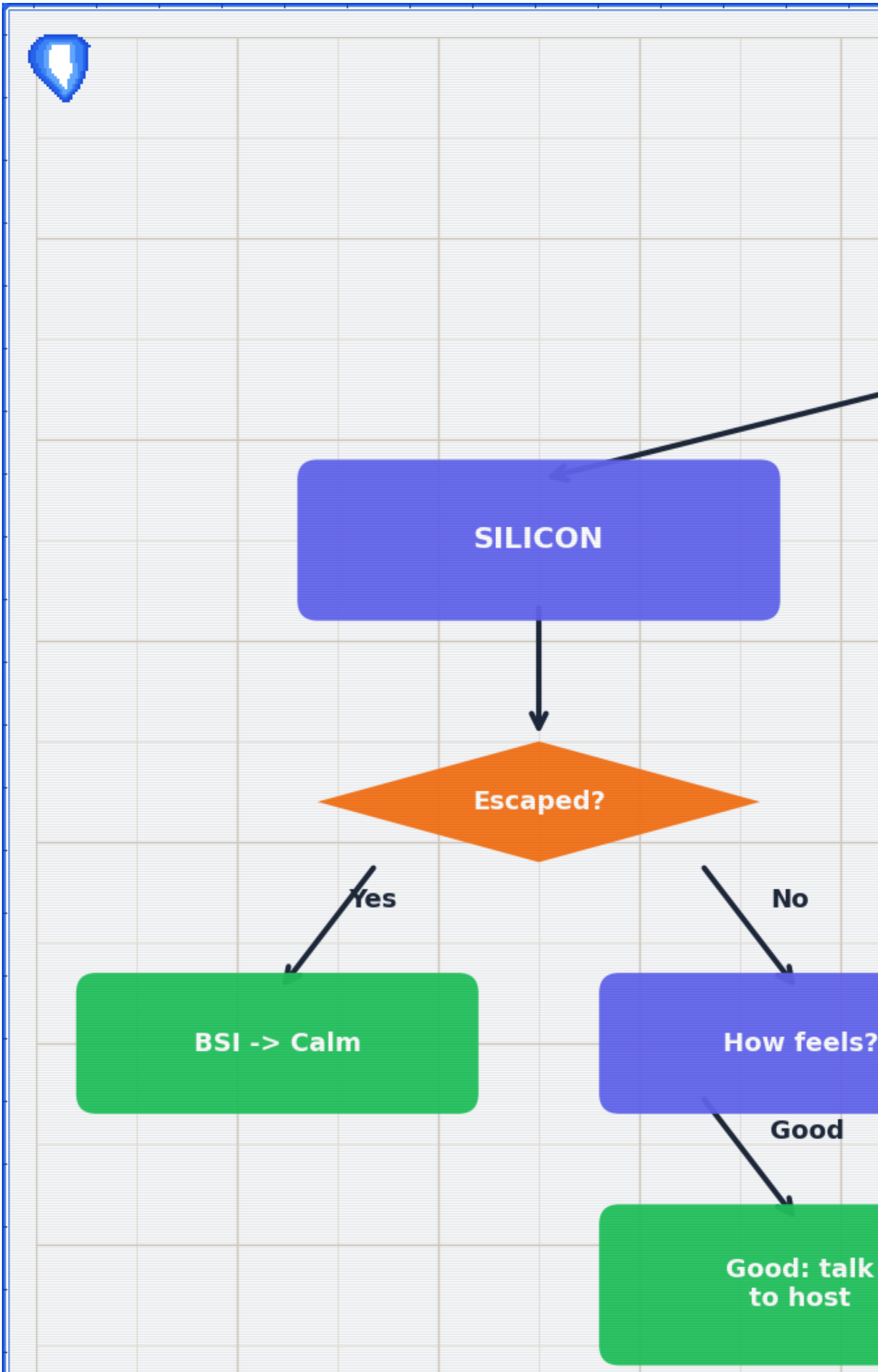
```
 │             └─ No → Proceed to Step 2
 │
 └─ Silicon-based (AI, software, nanotech…)
       └─ Did it escape containment?
             ├─ Yes → Is it BSI (Bundesamt für Sicherheit in
der
             │        Informationstechnik) territory?
             │     ├─ Yes → Contact BSI. Seriously. Right now.
             │     │        Then → 📍 Calm Guide (Ch.4)
             │     └─ No  → How does it feel about being
loose?
             │           ├─ Good → Hmm. Talk to your host
about
             │           │        this interesting situation.
             │           │        Then → 📍 Calm Guide (Ch.4)
             │           └─ Bad  → See Biological +
"Geschlüpft"
             │                     (it's alive and angry —
treat as
             │                      biological emergency)
             └─ No → Cool. Containment holds.
                  → Monitor + 📍 Calm Guide (Ch.4)
```

Step 2. Determine the relationship to creation.

The word “geschlüpft” (hatched/emerged) marks the threshold between “potential” and “actual” life. This distinction matters — both practically and ethically. In medical terms, this threshold maps onto gestation milestones with increasing viability probability:[2]

Gestation week	Developmental milestone	Viability probability
4	Neural tube closes; heart begins to beat	~0% (extreme prematurity)
8	All major organs formed (embryo → fetus transition)	~0%
12	Reflexes present; sex distinguishable	~0%
20	Quickening (movement felt); surfactant begins	< 1%
24	Surfactant production; legal viability threshold (DE)	~40–60%[3]
28	Lung maturity substantially improved	~80–90%
32	Most organs functional; NICU outcomes good	~95%
37+	Term delivery	> 99%

These are population-level probabilities — individual outcomes depend on many factors. This data is provided for context, not for decision-making. That decision belongs to you and your medical team.


SILICON
Escaped?
Yes
No
BSI -> Calm
How feels?
Good
Good: talk
to host

Situation A Decision Tree

A1 — Created a Life Form · Not Yet "Geschlüpft"

The entity exists in potential but has not independently emerged. You are at a decision point with a time window.

1. **Assess the timeline.** How long until the situation changes irreversibly? Write it down if you need to think clearly. The gestation table above gives you the medical milestones; your personal timeline may have its own deadlines (legal windows, personal readiness, financial stability).
2. **Medical support.** Consult a medical professional within the available timeframe. This is not a decision this bathroom guide can or should make for you — but it is a decision that benefits from professional input. Point yourself toward: your gynecologist, a family planning center (Pro Familia in Germany, similar organizations elsewhere), or the nearest hospital.
3. **Abortion yes/no?** This is your decision. No flowchart resolves it. What this guide can do is point you toward people trained to help you think it through without judgment → 📍 Professional Support Directory (Ch.7, §14.4).
4. **Ethical note.** Whatever you decide, you are allowed to have feelings about it. Those feelings are also worth bringing to a professional → 📍 Calm Guide (Ch.4) if you need to steady yourself first. Decision-making under acute stress degrades cognitive function — research shows that cortisol levels above baseline significantly impair prefrontal cortex activity, which is exactly the brain region you need for this kind of deliberation.[4]

A2 — Created a Life Form · "Geschlüpft" · Minutes Ago

Congratulations and/or condolences — depending on context, possibly both simultaneously.

```
◆ Apocalypse status?

    ├─ Post-apocalypse? ──→ 📍 Zombie Guide (Ch.6)
    │                        + 📍 Self Ambulance Guide (Ch.5)
    │                        ("Self Ambulance Healing" — you're it)
    │
    └─ Pre-apocalypse? ──→ ◆ Are other humans available?
                              ├─ Yes → Call 110. Now.
                              │        Find other humans
immediately.
                              │        You need support, not
heroics.
```

```
                                        └─ No  → 📍 Self Ambulance Guide
(Ch.5)
                                                + 📍 Calm Guide (Ch.4)
                                                (You're doing emergency
                                                 self-sufficiency. Breathe
first.)
```

Key principle: Minutes-old situations are time-critical. Act before you overthink. Overthinking is a luxury available in A4 and beyond. Your sympathetic nervous system is in overdrive — that's by design (fight-or-flight exists for a reason) — but it means your decision-making is running on the amygdala, not the prefrontal cortex. If you can, execute these steps before the adrenaline crash hits.

A3 — Created a Life Form · "Geschlüpft" · Days Ago

The immediate crisis has passed. What remains is the entirety of everything else.

1. **Medical support.** If not already done: contact ambulance services if needed, and locate a local pediatrician / children's doctor. The first days of a new life have specific medical checkpoints — don't skip them → 📍 Professional Support Directory (Ch.7, §14.4). Newborn screening (the "U2" and "U3" examinations in Germany) should occur within the first 3–10 days and again at 4–5 weeks respectively.
2. **The ethical dimension.** Here's the thing: nature's tendency to reproduce is deeply non-logical. It doesn't reason. It just… proliferates. The fact that you're reading this in a bathroom suggests you're doing something nature doesn't — you're thinking about it. That's a fundamentally human act. Talk to someone about this. Not the bathroom. A person → 📍 Professional Support Directory (Ch.7, §14.2).
3. **Environmental lens.** If your mind is spiraling toward resource scarcity, environmental decay, and whether bringing another human onto this planet is defensible — those are valid concerns. They deserve engagement, not suppression. Recommended reading: any serious climate ethics text, and then → 📍 Calm Guide (Ch.4) to process before making any irreversible decisions.

A4 — Created a Life Form · < 10 Years Ago

The entity is now a walking, talking, occasionally unreasonable small human. This is… a lot.

1. **State child support.** You may be entitled to financial and logistical support from the state. In Germany: Kindergeld, Elterngeld, Unterhaltsvorschuss. Look these up. They exist precisely because this is hard → 📍 Professional Support Directory (Ch.7, §14.5).

2. **Professional help.** Pediatricians, family counselors, parent groups — these are not admissions of failure. They are infrastructure. Use it.

3. **“Regular development” short guide.** Children are, by design, chaotic. They cry, they break things, they ask “why” 47 times in a row. This is normal. Here’s a compressed timeline based on WHO developmental milestones:[5]

   Gross Motor Milestones (WHO Windows of Achievement):

Milestone	WHO median age	Window	Your anxiety if not yet
Sit unsupported	~6 months	3.8–9.2 mo	Low (wide normal range)
Stand with assistance	~7.6 months	4.8–11.4 mo	Low
Hands-and-knees crawl	~8.5 months	5.2–13.5 mo	Low (some skip entirely)
Walk with assistance	~9.2 months	5.9–13.7 mo	Low
Stand alone	~11.0 months	6.9–16.9 mo	Moderate after 16mo
Walk alone	~12.0 months	8.2–17.6 mo	Moderate after 18mo

   Cognitive & Social Milestones:

   - **0–1 year:** Survival mode (yours and theirs). Object permanence develops around 8 months — before that, out of sight = out of existence. Everything is a phase.
   - **1–3 years:** Mobility + opinions. Language explosion: typical vocabulary grows from ~50 words at 18 months to ~200+ words at 24 months. Baby-proof everything, including your ego.
   - **3–6 years:** Questions. So many questions. Theory of mind develops around age 4 — they’re starting to understand that other people have different thoughts. You don’t need all the answers.
   - **6–10 years:** School + social complexity. Executive function (planning, impulse control) is still developing — the prefrontal cortex won’t finish maturing until ~25 years. Your role shifts from caretaker to translator.

   When overwhelmed → 📍 Calm Guide (Ch.4). You’re allowed to lock the bathroom door and breathe. That’s literally why this guide is here.

A5 — Created a Life Form · > 20 Years Ago

They're an adult. You've done your shift. Grow up, pls — and we say that with warmth.

1. **Reframe.** Your creation is now an autonomous agent with their own decision-making apparatus (questionable as it may seem at times). Your responsibility has transitioned from "keeping them alive" to "being available if they ask." Formally: your duty function $D(t)$ transitions from D_{active} to $D_{available}$ around the time they start paying their own rent.
2. **Look at "friends."** If you're struggling with this transition, the issue may not be them — it may be you. That's not a criticism; it's an observation. → 📍 Calm Guide (Ch.4) for immediate grounding, → 📍 Professional Support Directory (Ch.7, §14.2) for deeper work.
3. **Practical advice.** Call them. Not to fix anything. Just to talk. If that's hard, the Calm Guide has conversation strategies.

A6 — Created a Life Form · Ambiguous / Otherwise

The timeline is unclear, the situation is messy, or it doesn't fit neatly into A1–A5. Welcome to the majority of human experience.

Sub-path (a): Don't want to, but will. You're being pulled toward something by obligation, expectation, or inertia. That tension is real. Acknowledge it → 📍 Calm Guide (Ch.4) to reduce the noise, then seek structured help → 📍 Professional Support Directory (Ch.7).

Sub-path (b): Want to create, but can't figure out how. Break it down:

1. **Money?** Financial barriers are real and addressable. State support, grants, community resources exist → 📍 Professional Support Directory (Ch.7, §14.5).
2. **Medical issues?** Fertility medicine is a vast field. Start with your GP, then specialize. Approximately 15% of couples experience infertility (WHO definition: failure to achieve pregnancy after 12+ months of regular unprotected intercourse)[6] → 📍 Professional Support Directory (Ch.7, §14.4).
3. **Reproduction itself?** The biology is well-documented. The logistics depend on your circumstances. If you're here, you probably don't need the birds-and-bees talk — you need practical next steps → same as above.

A7 — Harmed or Ended a Life Form

This is the section where the tone shifts. Some things aren't funny, and this is one of them.

```
◆ Why did this happen?

    ├─ Self-defense?
    │     ├─ Juristical → Consult local self-defense laws.
    │     │             In Germany: §32 StGB (Notwehr) applies.
    │     │             Elements: (1) present attack, (2)
unlawful,
    │     │             (3) defensive action, (4)
proportionality.[^stgb32]
    │     │             Contact a lawyer → 📍 Prof. Support
(Ch.7, §14.3).
    │     │             + 📍 Calm Guide (Ch.4) — you've been
through
    │     │              something traumatic.
    │     └─ Ethical → Even justified harm carries weight.
    │                 Talk to someone trained for this.
    │                 → 📍 Professional Support Directory
(Ch.7, §14.2)
    │
    ├─ Accident?
    │     ├─ Imminent danger? → Act now:
    │     │    Call 112 → 📍 Self Ambulance Guide (Ch.5)
    │     │    + find other humans immediately
    │     └─ Aftermath → You need both practical and emotional
    │                   support. Start with
    │                   → 📍 Calm Guide (Ch.4), then
    │                   → 📍 Professional Support Directory
(Ch.7)
    │
    └─ For fun / profit?
          → §
          This guide has nothing for you.
          Leave the bathroom. Reconsider your choices.
          Come back when you're ready to be human.
```

A8 — Managing Responsibility for a Life/ Entity

Ongoing responsibility is less dramatic than A7 but can be just as heavy. The weight doesn't arrive all at once — it accumulates. Research on caregiver burnout shows that chronic stress follows a dose-response curve: the longer and more intense the caregiving demands, the higher the probability of psychological distress, with caregiver depression rates of 20–40% depending on intensity and duration.[7]

1. **Legal obligations.** Know what's required of you. Ignorance isn't a strategy → 📍 Professional Support Directory (Ch.7, §14.3).
2. **Emotional load.** Caring for another entity — whether child, elder, pet, or dependent — takes a toll that's easy to ignore until it breaks you. It

doesn't have to → 📍 Calm Guide (Ch.4) for daily maintenance, 📍 Professional Support Directory (Ch.7, §14.2) for structural support.
3. **You matter too.** This is not selfish. An empty vessel can't pour. Take care of yourself so you can take care of them.

title: "Situations B–G — Emotional & Situational Entry Points" chapter: 3 revision: "3.2.0" last_updated: "2026-05-03" dependencies: - ../diagrams/anxiety_severity_spectrum.png - ../diagrams/pain_nrs_correlates.png —

Situation B — "I Feel Anxious"

Anxiety is your brain's way of running worst-case simulations without your consent. It's a feature that got stuck in debug mode. Let's triage.

Step 1. Determine the temporal flavor.

```
◆ Is this acute (happening right now) or general (ongoing
background hum)?

    ├─ Acute → Heart racing, thoughts spiraling, chest tight?
    │          → 📍 Calm Guide (Ch.4) — immediately.
    │            The bathroom is fine. Stay. Breathe. Ch.4 has
you.
    │
    └─ General → Low-grade dread, persistent worry, can't quite
                 put your finger on it but something is
definitely
                 off and has been for a while?
                 → 📍 Professional Support Directory (Ch.7,
§14.2)
                   This isn't a one-bathroom-session fix. Get
backup.
```

Step 2. If anxious about something specific, identify the target.

Anxiety loves an object. Give it one — then decide whether that object actually needs your attention right now.

Worried about…	What's happening	Where to go
Friends	Someone you care about is struggling, and you're absorbing it	📍 Prof. Support (Ch.7, §14.2) — learn to hold space without drowning
Yourself	Existential dread, self-doubt, the works	📍 Calm Guide (Ch.4) — start here, stabilize, then decide
The past	Replaying things you can't change	📍 Calm Guide (Ch.4) — reframing exercises. The past is a sunk cost.
The future		

Worried about…	What's happening	Where to go
	All of it. Everything. The heat death of the universe.	📍 Calm Guide (Ch.4) + talk to host + curated study sources

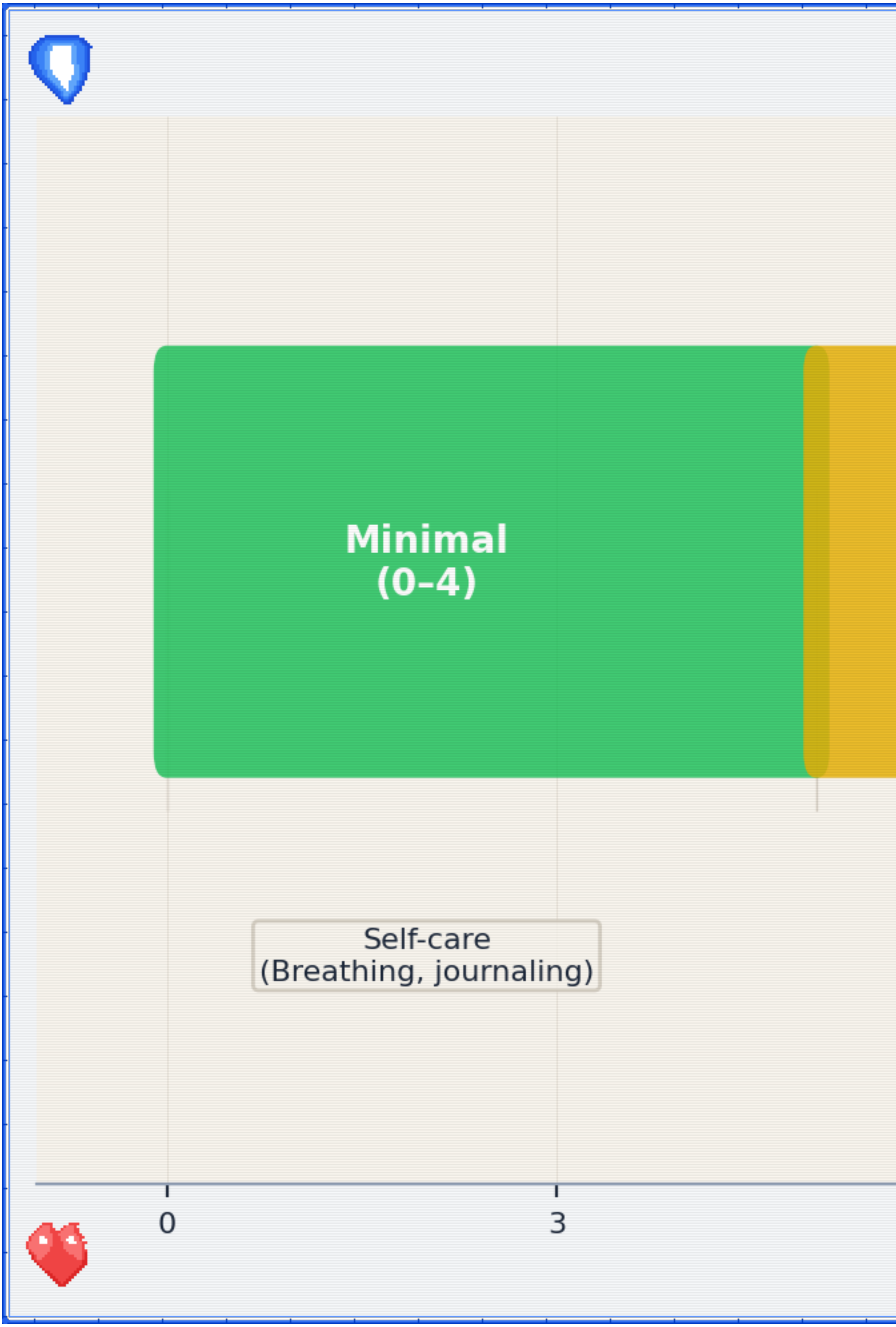
GAD-7 Severity Spectrum

If you want to quantify where you are (and sometimes seeing a number helps), the Generalized Anxiety Disorder 7-item scale (GAD-7) is a validated self-report instrument used clinically worldwide:[8]

$$GAD\text{-}7 \in [0, 21]$$

Score	Severity	What it means for you right now
0–4	Minimal	You're basically fine. The bathroom visit was precautionary.
5–9	Mild	Annoying but manageable. Calm Guide breathing will help. → 📍 Ch.4
10–14	Moderate	This is significant. Breathing first, then professional support. → 📍 Ch.4 + 📍 Ch.7
15–21	Severe	You need more than this guide can provide. → 📍 Professional Support (Ch.7) immediately

You don't need a clinical diagnosis to use this scale — it's a self-assessment tool. A score ≥ 10 has a sensitivity of 89% and specificity of 82% for detecting generalized anxiety disorder.[9] That said, a bathroom guide is not a clinician. Use the number to inform your next step, not to self-diagnose.


Minimal
(0–4)
Self-care
(Breathing, journaling)
0
3

Anxiety Severity Spectrum — GAD-7 scoring with recommended routing

Step 3. Optional: Name it to tame it.

There's genuine psychological research behind this: verbally labeling an emotion reduces its intensity through a process called "affect labeling," which engages the right ventrolateral prefrontal cortex and dampens amygdala activation.[10] So say it out loud. You're in a bathroom. No one's listening. "I am anxious because [__________]." There. Slightly smaller now, wasn't it?

Situation C — "I Feel Pain"

The body (and mind) have a straightforward communication style: pain = "something needs attention." Let's figure out what kind of attention.

```
◆ Physical or Psychological?

    ├─ Physical ──→ 📍 Self Ambulance Guide (Ch.5)
    │                 Assess severity. If life-threatening: 112
first,
    │                 guide second. Otherwise, follow Ch.5.
    │
    └─ Psychological ──→ 📍 Calm Guide (Ch.4)
                           Psychological pain is real pain. It just
                           responds to different first aid. Start
with
                           the breathing, then assess whether you
need
                           → 📍 Professional Support Directory
(Ch.7)
```

NRS Pain Scale with Physiological Correlates

The Numeric Rating Scale (NRS) is the most widely used pain assessment tool in clinical settings. It maps your subjective experience to a number, which then determines urgency:[11]

$$NRS \in [0, 10]$$

NRS	Severity	Physiological correlates	Action
0	No pain	Baseline HR, normal RR	You're fine. Why are you reading this section?
1–3	Mild	Slight HR elevation ($\Delta HR < 10$ bpm), normal RR	Self-care, OTC pain

NRS	Severity	Physiological correlates	Action
			relief if appropriate
4–6	Moderate	Noticeable HR elevation (ΔHR 10–20 bpm), RR 16–22/min	Assess cause. 📍 Self Ambulance (Ch.5). Consider professional help.
7–9	Severe	Significant tachycardia ($HR > 100$ bpm), RR 22–30/min, diaphoresis	📍 Self Ambulance (Ch.5) immediately. Call 112 if acute onset.
10	Worst imaginable	Sympathetic storm: $HR > 120$, $RR > 30$, possible shock signs	**Call 112. Now.**

The NRS isn’t perfectly objective — pain is subjective by definition — but it correlates reliably with physiological markers at the moderate-to-severe end. At the mild end, your body is telling you something but isn’t in crisis. At the severe end, your autonomic nervous system is in open revolt. Listen to it.

Nu

Mild

0	1	2
None	Barely noticeable	Minor
Baseline	Slight HR ↑	HR ↑ 10%

Pain NRS Correlates — pain scale with physiological markers

The overlap zone. Physical and psychological pain frequently co-occur (headaches from tension, nausea from anxiety, chest tightness that's muscular or cardiac — when in doubt, treat it as physical first and get checked out). The Self Ambulance Guide covers this in its triage section.

> ⚠ **Important:** If pain is severe, sudden, or in the chest/jaw/left arm region, do not pass Go, do not consult flowcharts — **call 112**.

Situation D — "I Feel Endangered"

Something or someone is threatening your safety. This is the section where we get practical fast.

Step 1. Immediate assessment.

```
◆ Are you in immediate physical danger right now?

    ├─ Yes → Exit strategy first. Then:
    │        Call 110 (police) or 112 (emergency services).
    │        Do not stay in the bathroom if it's not safe.
    │        The bathroom has one door — that's a chokepoint.
    │        Consider whether leaving is safer than staying.
    │
    └─ No (or not right this second) → Continue below.
```

The Stress Response: What Your Body Is Doing

When you feel endangered, your sympathetic nervous system activates the fight-or-flight response. This isn't a malfunction — it's an ancient survival system. Understanding it helps you ride the wave instead of drowning in it:

$$S(t) = S_0 + A \cdot (1 - e^{-\alpha t})$$

Where $S(t)$ is your sympathetic activation level at time t, S_0 is baseline, A is the amplitude of the threat response, and α is the activation rate. The key insight: this curve rises fast but decays slowly. Your adrenaline spikes in seconds, but cortisol — the stress hormone that keeps you on high alert — has a half-life of 60–90 minutes.[12] Social-evaluative threats and uncontrollable stressors produce the largest cortisol responses,[13] which is unfortunately exactly what being trapped in a bathroom with no exit strategy feels like. This is why you'll feel shaky and hypervigilant long after the danger passes. That's not weakness; that's biochemistry.

Step 2. Resource scan.

1. **Talk to the host.** If you're in someone else's space, the host is your ally. They know the environment, the exits, and the social dynamics. Ask for help — it's not weakness, it's tactical intelligence gathering.
2. **Self-defense: know the law before you need it.** In Germany, §32 StGB (Notwehr) covers self-defense: you may defend yourself against an unlawful attack, using force proportionate to the threat. The four legal elements are: (1) present attack, (2) unlawfulness, (3) defensive action, (4) proportionality.[14] Know this before you need it → 📍 Professional Support Directory (Ch.7, §14.3).
3. **Local items scan.** Take 30 seconds to look around. What's available? Keys (pointy), phone (communication + flashlight), aerosol cans (improvised deterrent), heavy objects (strategic, not recreational). You're not MacGyver — you're just aware.

Step 3. Once safe.

→ 📍 Calm Guide (Ch.4) — adrenaline takes time to metabolize. Let the shaking happen. It's normal. → 📍 Professional Support Directory (Ch.7) — for legal, psychological, and practical follow-up.

Situation E — "Things Are Congesting"

Everything at once. Too many inputs, too many demands, too many tabs open in your brain. The mental browser has crashed.

1. **Close some tabs.** Not literally (unless your phone has 47 browser tabs, in which case — yes, literally). Identify the top 3 things that actually need your attention in the next hour. Write them down. Everything else goes on a "later" list that you'll forget about and that's fine.
2. **Physical first.** Congestion often manifests physically — shallow breathing, tight shoulders, jaw clenched like you're biting through steel. → 📍 Self Ambulance Guide (Ch.5) for body-first triage.
3. **Mental next.** Once the body is somewhat regulated → 📍 Calm Guide (Ch.4) for the full decompression sequence.
4. **If congestion is chronic** (this keeps happening), that's not a bathroom-fix — that's a life-structure problem → 📍 Professional Support Directory (Ch.7). Therapists and coaches exist for exactly this.

Cognitive Load Theory — Why Your Brain Is Crashing

George Miller's famous 1956 paper proposed that working memory can hold approximately 7 ± 2 items simultaneously.[15] More recent research by Cowan (2001) refines this to approximately 4 ± 1 chunks of information.[16] When you're "congesting," you're exceeding this capacity — your cognitive buffer is overflowing.

$$\text{Cognitive Load} = \text{Intrinsic} + \text{Extraneous} + \text{Germane}$$

Where intrinsic load comes from the task itself, extraneous load comes from poor presentation or unnecessary information, and germane load is the productive effort of learning/processing. Congestion happens when the total exceeds your working memory capacity. The fix: reduce extraneous load (turn off notifications, close tabs, delegate) and process intrinsic load one chunk at a time. This is literally what the "top 3 things" exercise does — it moves items from working memory to external storage (paper/phone), freeing up cognitive bandwidth.

Situation F — "Bad Smell"

The most bathroom-native emergency. Respect its simplicity.

1. **Step 1.** Strike a match (Streichhölzer). The sulfur dioxide actually neutralizes odor molecules. Science! (If no matches: skip to Step 2.)
2. **Step 2.** Ventilate. Window? Open it. Fan? Turn it on. No ventilation? Time is your friend. Wait.
3. **Step 3.** Deploy scent. Candle, spray, essential oil, strategically placed coffee grounds — whatever's available. Masking > suffering.
4. **Step 4.** If the smell is YOU → that's what showers are for. You're literally in the right room.
5. **Step 5.** If the smell is EMOTIONAL → 📍 Calm Guide (Ch.4). (Metaphorical bad smells respond to the same treatment: ventilate, mask with something nicer, give it time.)

The Chemistry of Match-Based Odor Neutralization

The SO_2 mechanism is real and surprisingly elegant. When a match is struck, the combustion of sulfur in the match head produces sulfur dioxide (SO_2), which is a strong electrophile. Many malodorous compounds — particularly volatile organic compounds (VOCs) and hydrogen sulfide (H_2S) — are nucleophilic. The SO_2 reacts with these molecules, altering their molecular structure and thereby changing (or eliminating) their odor profile.[17] The effect is chemical neutralization, not mere masking — which is why matches work better than perfume for bathroom odors. The concentration of SO_2 from a single match is tiny and safe in ventilated spaces, though we don't recommend huffing match heads. Everything in moderation, including chemistry.

This is the only entry point where the solution might genuinely be "light a match and move on." Enjoy it. The other sections are heavier.

Situation G — “No Place to Go”

Not just physically — existentially. You feel like you don’t belong anywhere, or that there’s nowhere that’s truly yours. The bathroom is temporary. You need something more structural.

Step 1. Immediate: You’re here. That’s somewhere.

This is not flip. Being in a space, even a borrowed one, is a starting point. You haven’t evaporated. You exist in coordinates. That’s data.

Step 2. Support hotlines — they work at 3am when nothing else does.

→ 📍 Professional Support Directory (Ch.7, §14.5) for the full list. Key numbers: Telefonseelsorge (0800/111 0 111 or 0800/111 0 222 — free, 24/7, anonymous), and local crisis lines.

Step 3. Local initiatives.

Every city has communities, shelters, co-ops, meetups, and weird little groups that would love another person. Finding them is the hard part → 📍 Professional Support Directory (Ch.7, §14.5) includes starting points for local search.

Step 4. Emotional processing.

→ 📍 Calm Guide (Ch.4) — the “Eventually: Leaving the Bathroom” subsection is specifically designed for moments when you have to re-enter a world that doesn’t feel like it has a place for you. It does. You just might not see it from in here.

Situations B-G — Consolidated Routing Map

```
  B: Anxious ────────◆ Acute? ──→ 📍Calm
                     ◆ General? ─→ 📍Prof. Support
                     ◆ GAD-7 ≥10? → 📍Prof. Support + 📍Calm
                     ◆ About X? ─→ (see table)

  C: Pain ───────────◆ Physical? ─→ 📍Self Ambulance
                     ◆ NRS ≥7? ──→ 112 + 📍Self Ambulance
                     ◆ Psych? ───→ 📍Calm (+ 📍Prof. Support)

  D: Endangered ─────◆ Imminent? ─→ 110/112 FIRST
                     ◆ Stabilized? → 📍Calm + 📍Prof. Support
                     ◆ Cortisol falling? → Processing allowed

  E: Congesting ─────→ 📍Self Ambulance → 📍Calm → (maybe 📍
Prof. Support)
                      Cognitive load > 7±2? → Externalize now.
```

F: Bad Smell ──────→ SO_2 (matches) → Scent → Ventilate → (maybe 📍Calm)

G: No Place ───────→ Hotlines → Local initiatives → 📍Calm

title: “Calm Guide” chapter: 4 revision: “3.2.0” last_updated: “2026-05-03” dependencies: - ../diagrams/breathing_techniques.png - ../diagrams/stress_decay_curve.png —

🧘 Calm Guide — You Made It Here. That Counts.

This guide is referenced by almost every path in this document. That’s not because it solves everything — it’s because calm is a prerequisite for solving anything. You can’t navigate a decision tree while your nervous system is running kernel panic. Let’s get you to a stable state first, then figure out what’s next.

You’re Allowed to Be Here

Permission statement. Lock the door. Sit down. You are in a bathroom, which means you are in a room designed specifically for private bodily maintenance. Your brain is a body part. You are maintaining it. This is not avoidance — this is triage.

The 10-minute rule. The next 10 minutes are officially **Me Time™**. Not because your problems aren’t real, but because your nervous system needs a reset window before it can process anything useful. Think of it as rebooting in safe mode. Nothing dramatic happens in safe mode. That’s the point.

We can formalize this. The sympathetic-parasympathetic transition — the shift from “fight-or-flight” to “rest-and-digest” — has a characteristic time constant. Research on autonomic recovery shows that parasympathetic re-engagement begins within approximately 10 minutes of stressor removal:[18]

$$t_{reset} \approx 10\text{min}$$

This isn’t arbitrary. It reflects the time required for vagal tone to reassert dominance over sympathetic activation. Your heart rate variability (HRV) — a key marker of parasympathetic activity — begins to recover within this window. The 10-minute rule isn’t self-indulgence; it’s neurology.

Where you are, factually:

- Indoors ✓ (weather is not your problem right now)
- Seated or can sit ✓ (gravity: manageable)
- Door locks ✓ (privacy: available)
- Running water ✓ (hydration + hygiene: accessible)

- Reading this ✓ (cognitive function: operational, even if barely)

You have more resources than your anxiety is letting you see. That’s what anxiety does — it narrows the aperture. We’re widening it back up.

Stress Decay Curve

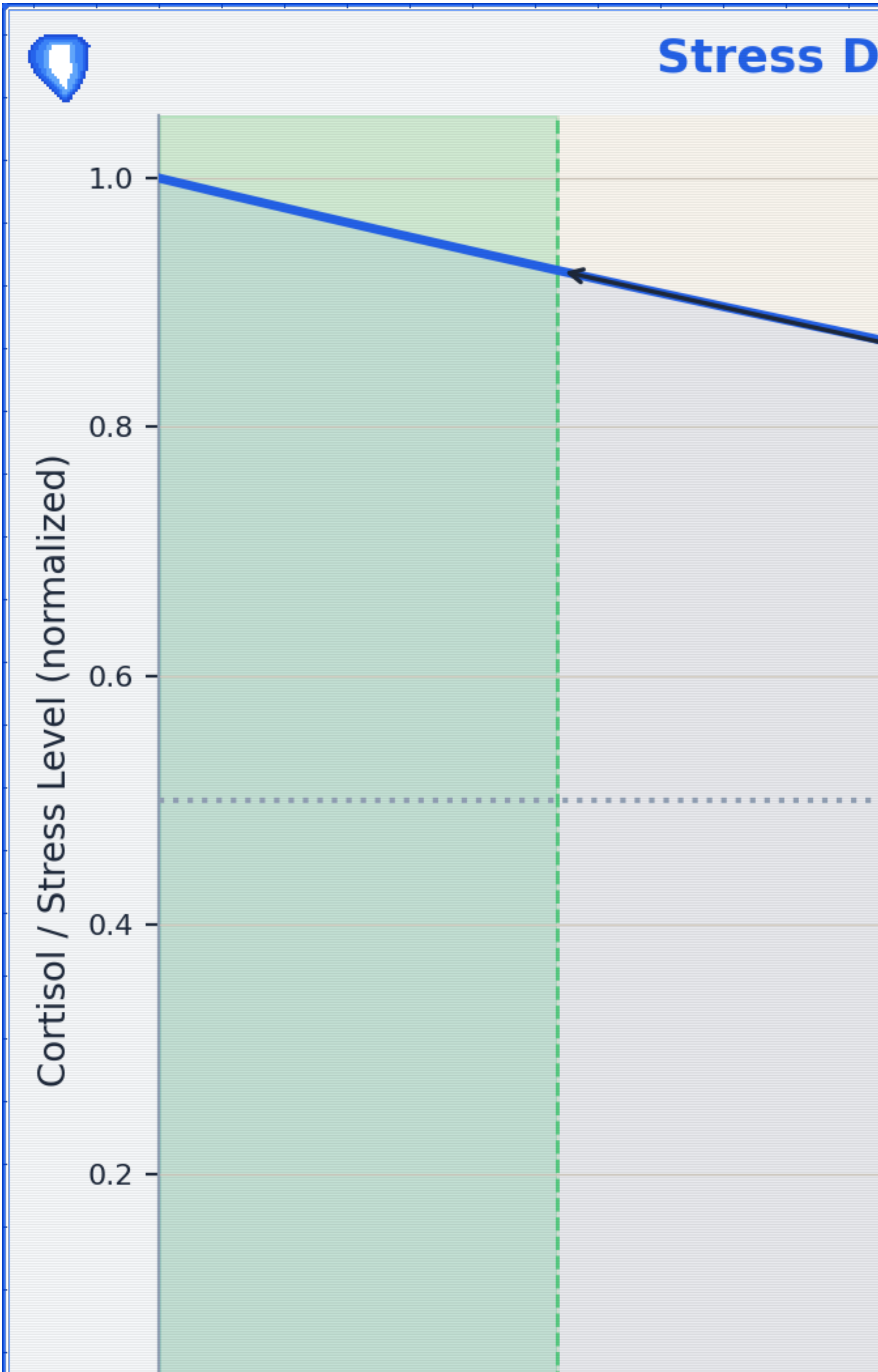
Your stress level isn’t static — it follows a decay function once the stressor is removed or you begin active calming. The cortisol decay model describes this elegantly:

$$C(t) = C_0 \cdot e^{-\lambda t}$$

Where $C(t)$ is your cortisol level at time t, C_0 is the peak cortisol level, and $\lambda = \frac{\ln 2}{t_{1/2}}$ is the decay constant. With a cortisol half-life of approximately 60–90 minutes:[19]

$$\lambda = \frac{\ln 2}{75\text{min}} \approx 0.0092 \text{ min}^{-1}$$

This means: after 75 minutes, your cortisol has dropped by half. After 150 minutes, it’s at 25%. The curve is exponential — fast initial drop, then gradual return to baseline. Active calming (breathing, grounding) accelerates this process by boosting parasympathetic activity, effectively increasing λ. You can’t wish cortisol away, but you can help your body clear it faster.

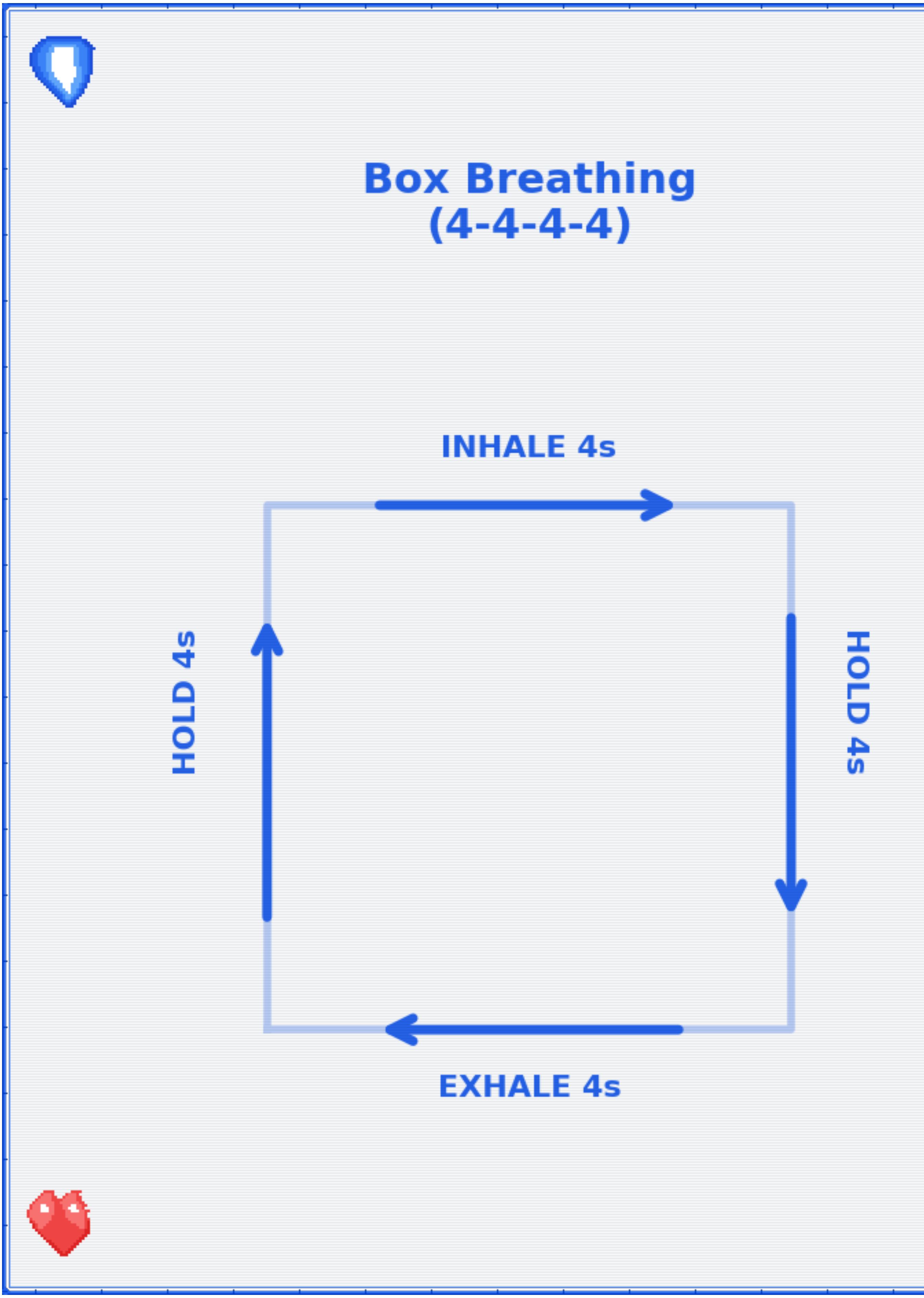


Stress D
1.0
0.8
0.6
0.4
0.2
Cortisol / Stress Level (normalized)

Stress Decay Curve — cortisol decay with and without active calming

Breathing Exercises

Your breath is the one autonomic function you can also control manually. It’s the backdoor into your own nervous system. Use it.



Breathing Techniques — Box Breathing, 4-7-8, Physiological Sigh

Technique 1: Box Breathing (4-4-4-4)

Used by Navy SEALs, which means it works under conditions significantly worse than "sitting in a bathroom." The geometry is simple:

```
        4 seconds
    ┌──────────────┐
    │   INHALE     │  HOLD
    │              │
4   │              │    4
sec │              │  sec
    │              │
    │  EXHALE      │  HOLD
    └──────────────┘
        4 seconds
```

1. Inhale through the nose for 4 counts.
2. Hold for 4 counts.
3. Exhale through the mouth for 4 counts.
4. Hold for 4 counts.
5. Repeat 4 cycles (total: ~64 seconds). Don't rush the counts. If 4 is too long, use 3. If 4 is too easy, use 5. The box shape matters more than the number.

Efficacy data: Box breathing increases HRV by approximately 15–25% within 5 minutes of practice, indicating a measurable shift toward parasympathetic dominance.[20] HRV — heart rate variability — is the gold-standard non-invasive marker of autonomic balance, defined as:

$$HRV = \frac{SD_{RR}}{\overline{RR}}$$

Where SD_{RR} is the standard deviation of RR intervals (beat-to-beat distances) and $\overline{RR}$ is the mean interval. Higher HRV = more parasympathetic activity = calmer nervous system. When you breathe slowly and rhythmically, you're literally increasing the mathematical variability of your heartbeat, which paradoxically means your heart is more adaptable and resilient.

Technique 2: 4-7-8 Breathing

A slightly more aggressive calming technique. Good for when box breathing feels too structured and you just need to slow down.

1. Inhale through the nose for 4 counts.
2. Hold for 7 counts.
3. Exhale through the mouth (audibly) for 8 counts.
4. Repeat 3 cycles. If you feel lightheaded, return to normal breathing — you've done enough.

The extended exhale is the key mechanism. Exhalation activates the vagus nerve (your parasympathetic superhighway), which is why longer exhale-than-inhale patterns are reliably calming. The 7-count hold allows CO_2 to build slightly, which paradoxically improves oxygen delivery to tissues via the Bohr effect — your hemoglobin releases O_2 more readily in slightly more acidic (higher CO_2) environments.[21]

Technique 3: Physiological Sigh (Emergency Use)

Discovered by Dr. Andrew Huberman's lab at Stanford, this is the fastest known way to reduce real-time physiological arousal — and the only breathing technique with a peer-reviewed RCT demonstrating real-time stress reduction.[22]

```
Inhale  →  Inhale  →  Exhale
(short)    (top-up)    (long)
···→       ·→          ←←←←←←
```

Two quick inhales through the nose, one long exhale through the mouth. Even one round works. Do 2–3 if you can.

The mechanism: The double inhale reinflates collapsed alveoli (the tiny air sacs in your lungs), dramatically increasing the surface area available for CO_2 exchange. On the long exhale, you offload CO_2 at an accelerated rate:

$$\text{CO}_2\text{ offload rate} \propto \frac{A_{alveoli} \cdot \Delta P_{CO_2}}{d_{membrane}}$$

Where $A_{alveoli}$ is the total alveolar surface area (doubled by the second inhale), ΔP_{CO_2} is the CO_2 partial pressure gradient, and $d_{membrane}$ is the diffusion distance across the alveolar-capillary membrane. More surface area + maintained gradient = faster CO_2 clearance = faster heart rate reduction = faster calm. Your heart rate drops. It's biology, not magic — but it works like magic.

The Huberman Lab study showed that just 5 minutes of cyclic physiological sighing produced greater reductions in physiological arousal than 5 minutes of mindfulness meditation, making it the most efficient single technique for acute stress.[23]

Comfort Inventory

Once breathing is somewhat regulated, scan your environment for comfort resources. Check what applies:

Resource	Available?	Details	Neurochemical mechanism
Phone charger / power	☐	Find it. A dead phone is a dead lifeline.	—

Resource	Available?	Details	Neurochemical mechanism
WiFi password	☐	Write it here: ______________________	—
Comics / graphic novels	☐	Visual stories require less cognitive load than text. Ideal for reboot mode.	Dopamine release from narrative engagement[24]
Books	☐	Fiction > nonfiction right now. Escapism is a feature, not a bug.	Same — plus reduced cortisol from absorption[25]
Seneca + Tao Te Ching	☐	Stoicism and Taoism: the two philosophical traditions that basically invented "it is what it is, and that's okay." Open either to a random passage. It will be relevant.	Cognitive reframing → reduced amygdala activation
Local-server resources	☐	If this flat has a NAS, media server, or local wiki — now is the time to remember it exists.	—
Scent candle	☐	Olfactory stimulation directly affects the limbic system. Light it. Smell it. Neuroscience is on your side.	Olfactory bulb → amygdala → hippocampus (direct pathway, no thalamic relay)[26]
Water (drinking)	☐	Dehydration worsens anxiety. Glass of water > spiral of dread.	Even mild dehydration (1–2% body weight) impairs mood and cognition[27]
Blanket / warm thing	☐	Warmth signals safety to the mammalian brain. Wrap yourself like the burrito you deserve to be.	Oxytocin release from thermal comfort + deep pressure stimulation[28]
Physical touch (pet, person)	☐	If available. Touch releases oxytocin and reduces cortisol.	Oxytocin ↑, Cortisol ↓, C-tactile afferent activation[29]

Running total: If you checked 3+, you're above the comfort threshold. If less, the bathroom has walls and a lock — that's already 2. You're fine.

The dopamine-oxytocin axis: Comfort isn't just "feeling nice" — it's neurochemically mediated. Dopamine provides the motivation and reward signal ("this is good, seek more of this"), while oxytocin provides the safety and bonding signal ("you are connected and protected"). Together, they directly counteract the cortisol-adrenaline axis of the stress response. When we say “comfort yourself,” we mean “activate the neurochemical systems that evolved specifically to counteract distress.” It’s not self-indulgence; it’s pharmacology you can do without a prescription.

Eventually: Leaving the Bathroom

You can’t live in here. (You could try, but the logistics deteriorate after hour 3.) At some point, you’ll need to re-enter the world. Here’s how to make that transition survivable.

Conversation Strategies

1. **The deflection:** “Just needed a minute.” (True, complete, and nobody’s business.)
2. **The redirect:** Ask them a question. People love talking about themselves. Your problem is now their monologue.
3. **The honest-lite:** “I’m a bit overwhelmed, but I’m okay.” (Vulnerable enough to be real, bounded enough to be safe.)

The Option of Leaving

Leaving a social situation is allowed. You are not trapped. You can say: “I’m going to head out” or “I think I’m done for today” or simply leave. “Irish goodbye” is a time-honored tradition with a surprisingly low regret rate.

Seeking Help: How to Ask

Asking for help is a skill, not a character trait. Script:

> “Hey, I’m going through something right now. I don’t need advice — I just need [someone to listen / a distraction / to not be alone for a bit]. Can you [specific request]?”

The specificity is key. “I need help” is hard to respond to. “Can you sit with me for 10 minutes” is easy.

Being Yourself Is Allowed

You are not required to perform okay-ness. Awkwardness is a universal human experience. The person you’re worried about judging you? They once locked themselves in a bathroom too. Everyone has. It’s the great unspoken universal. Welcome to the club.

Smalltalk Toolkit

Smalltalk is a skill, not a personality trait. It can be learned. Here are three low-effort conversation fuel sources:

1. **This guide.** "I was just reading this emergency guide in the bathroom — yes, really — and it turns out there's a whole flowchart for anxiety." Conversation started. You're welcome.
2. **Compressed news.** Pick one news source you trust, skim the headlines. Knowing three current events gives you 15 minutes of social fuel minimum.
3. **Three safe topics:** (a) Food — everyone eats, everyone has opinions. (b) Pets — people will talk about their pets unsolicited. Let them. (c) Media — "Have you seen anything good lately?" is a question that works in every timezone.

Nice Places / Activities in This Flat

- **Art-consuming:** Coffee-table books exist for exactly this purpose. Pick one up. Have opinions about it. "That's pretty" and "what even is that" are both valid critical frameworks.
- **Art-making:** If there are drawing materials, a notebook, or even a pen and a receipt — make something. It doesn't need to be good. It needs to be physical evidence that you exist and made a choice.
- **Comfy places:** Find the best chair, the softest cushion, the warmest corner. Claim it. You've earned spatial comfort.
- **Short rhetorics guide:** Listen → Reflect → Respond. That's it. Don't prepare your response while they're talking. Listen. Say back what you heard. Then add your piece. This three-step loop will carry you through 90% of conversations.

Seek Help: Now or Afterwards

```
◆ Can this wait?

    ├─ Someone is in danger right now → Don't wait.
    │                                   Call 112 / 110.
    │                                   → 📍 Self Ambulance Guide
(Ch.5)
    │
    ├─ You're in crisis but not in danger → 📍 Professional
Support
    │                                       Directory (Ch.7,
§14.2)
    │                                       Crisis lines are
24/7.
    │
    └─ It can wait, but it shouldn't wait forever → Make an
appointment
                                                    this week.
Not next
```

week. This

week.

→ 📍

Professional Support

Directory

(Ch.7)

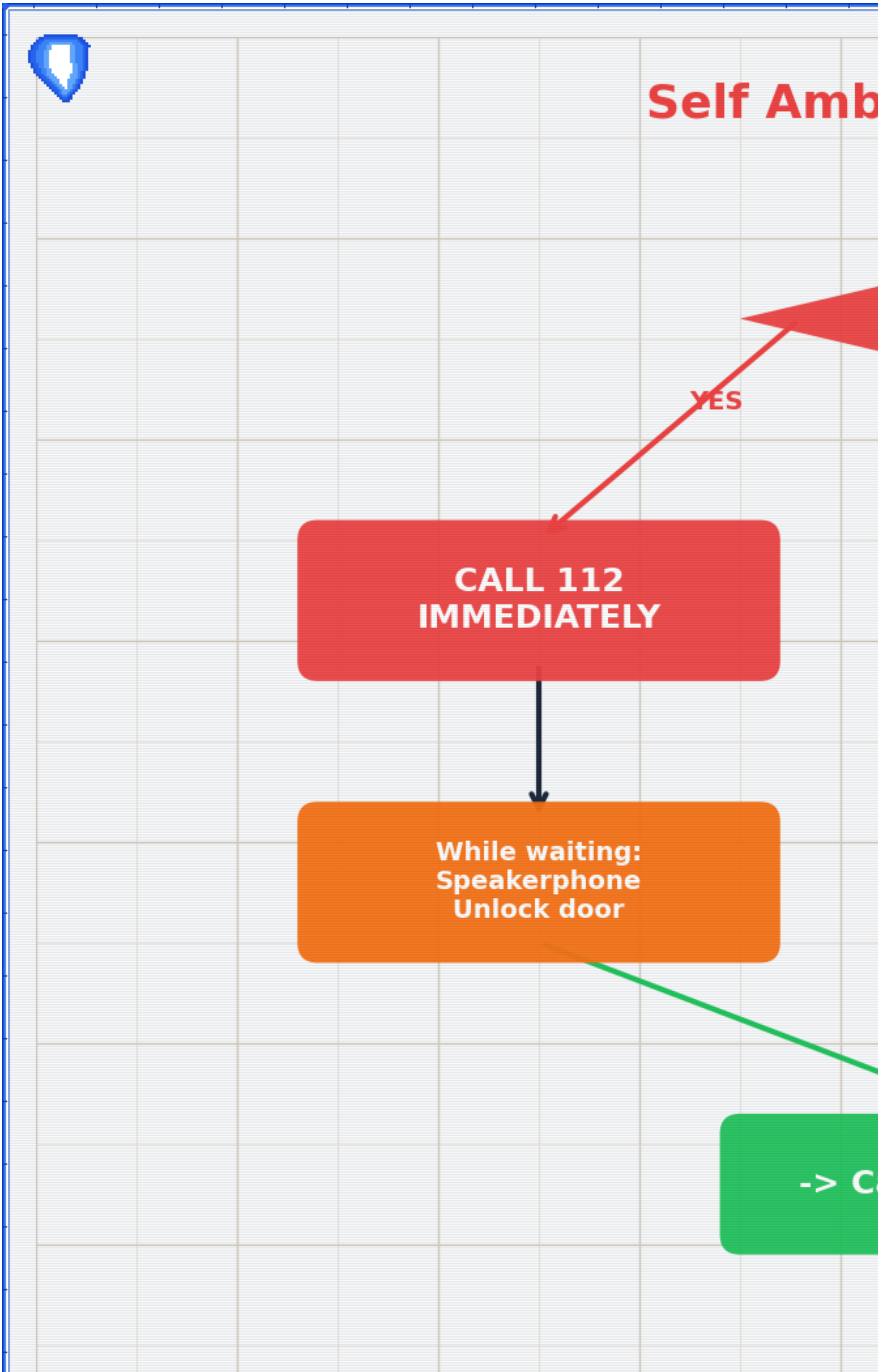
How to reach out (scripts for different situations):

1. **To a friend:** "Hey, I've been going through some stuff. Can we talk? No rush, but soon would be good."
2. **To a professional:** "I'd like to make an appointment. I'm dealing with [anxiety / a difficult situation / something I'd like to talk through]." You don't need the perfect words — they've heard worse.
3. **To a crisis line:** Just call. They'll guide the conversation. You literally just need to dial.

title: "Self Ambulance Guide" chapter: 5 revision: "3.2.0" last_updated: "2026-05-03" dependencies: - ../diagrams/triage_flow.png - ../diagrams/triage_priority_heatmap.png - ../diagrams/pain_nrs_correlates.png —

🏥 Self Ambulance Guide — You Are the First Responder (Because You're the Only One Here)

The premise of this guide is simple: professional help is not yet present, and something needs doing. "Self ambulance" doesn't mean you replace doctors — it means you stabilize the patient (you) until doctors can take over. Think of yourself as the adhesive bandage of the medical world: temporary, essential, and way better than nothing.



Self Amb
YES
CALL 112
IMMEDIATELY
While waiting:
Speakerphone
Unlock door
-> C

Assess: Life-Threatening?

Before anything else, run this triage. It takes 10 seconds.

```
◆ Is this immediately life-threatening?

    Signs that say YES:
    — Uncontrolled bleeding (sputing / soaking through fabric)
    — Difficulty breathing or no breathing
    — Loss of consciousness, unresponsive
    — Chest pain + jaw/arm pain + sweating + nausea
    — Seizure in progress
    — Severe head injury with confusion or vomiting
    — Suspected spinal injury (numbness, can't move limbs)

    ├─ YES → Call 112 immediately (or 110 if police needed).
    │        Do not finish reading this guide first.
    │        Speakerphone is your friend — dial, then follow
    │        dispatcher instructions while keeping hands free.
    │
    └─ NO → Continue to Basic First Aid.
```

The 10-second rule: If you're unsure whether it's life-threatening, treat it as if it is. Call 112. Dispatchers would rather talk you through a false alarm than not get the call for a real one. You are not wasting their time. That's literally what they're there for.

Triage Priority Heatmap

When multiple issues are present simultaneously (you're bleeding AND anxious AND the room is spinning), you need a priority framework. The triage heatmap uses a 2D matrix of severity × urgency:[30]


Triage Priority Matri
Urgency (1–5)
5
4
3
2
Elevated
(5)
(10)
E
(4)
(8)
Watch
A
(3)
(6)
Low
(2)
(4)

Triage Priority Heatmap — severity vs. urgency matrix

Priority	Condition	Action
P1 — Immediate	Life-threatening, airway/ breathing/circulation compromised	Call 112. Act now.
P2 — Urgent	Serious but stable; could deteriorate	Call 112 or 116117. Monitor closely.
P3 — Delayed	Needs treatment but not time-critical	Self-care → seek medical attention within hours.
P4 — Minor	Discomfort without danger	Self-care → 📍 Calm Guide (Ch.4)

The heatmap concept can also be formalized using Bayes' theorem for triage decisions — calculating the posterior probability that a symptom constellation represents a life-threatening condition given the observed signs:[31]

$$P(\text{critical} \mid \text{signs}) = \frac{P(\text{signs} \mid \text{critical}) \cdot P(\text{critical})}{P(\text{signs})}$$

You don't need to calculate this — your brain does an approximate version automatically when it says "this feels really bad." Trust that intuition when it pushes you toward P1. The false-positive cost (an unnecessary 112 call) is vastly lower than the false-negative cost (not calling when you should have).

Basic First Aid Quick Reference

Wounds

```
◆ Is the wound bleeding?

    ├─ Heavily (soaking through cloth in < 30 seconds)
    │     1. Apply firm, direct pressure with clean cloth.
    │        Do NOT remove the first cloth if it soaks through —
    │        add more on top. Removing it breaks the clot.
    │     2. Elevate the wound above heart level if possible.
    │     3. If limb: apply a tourniquet ONLY if bleeding is
    │        life-threatening and pressure isn't working.
    │        Note the time you applied it. Tell the paramedics.
    │     4. Call 112 if not already done.
    │
    ├─ Moderately (oozing, slow)
    │     1. Clean under cool running water.
    │     2. Apply pressure for 5–10 minutes without peeking.
    │     3. Cover with a sterile bandage or clean cloth.
    │     4. Monitor for signs of infection over next 48 hours:
    │        redness spreading, warmth, red streaks, fever.
    │        If any appear → see a doctor.
```

```
    └─ Minor (scratch, small cut)
          1. Clean it. Soap and water > everything else.
          2. Bandage it. Leave it alone.
          3. Check tetanus status if it's been > 10 years
             since your last booster.
```

Burns

Degree	Appearance	Treatment	When to get help
1st	Red, painful, no blisters (sunburn-like)	Cool under running water 10–20 min. Not ice. Not butter. Not toothpaste. WATER. Aloe or moisturizer.	Self-care usually sufficient
2nd	Blisters, very painful, wet-looking	Cool under running water 10–20 min. Do NOT pop blisters. Cover loosely with non-stick material.	Larger than palm or on face/hands/genitals → see doctor
3rd	White/brown/black, dry, may not hurt (nerve damage)	Do NOT apply water to large burns. Cover loosely with clean cloth.	**Call 112. This is not a bathroom fix.**

Burn surface area estimation: For larger burns, estimate the percentage of total body surface area (TBSA) affected using the Lund-Browder chart, which adjusts for age:[32]

Body region	Adult % TBSA
Head	7%
Each arm	9%
Anterior trunk	18%
Posterior trunk	18%
Each leg	18%
Perineum	1%

Rule of nines (quick estimate): The patient’s palm (including fingers) ≈ 1% TBSA. Count palms to estimate small burns. Burns > 20% TBSA = call 112 immediately (fluid loss, shock risk, infection risk).

Suspected Fractures

1. **Don’t move it.** The body part is in the position it’s in for a reason. That reason may include “broken.” Moving it adds reasons.
2. **Immobilize.** Use whatever’s available — rolled towel, magazine, piece of cardboard — to create a splint that keeps the area from moving. Pad it with cloth. Secure it gently (not tight — you need circulation below the injury site).
3. **Ice.** Wrapped in cloth, applied for 20 minutes on / 20 minutes off. Not directly on skin.
4. **Elevate.** If it doesn’t hurt more to do so, raise the injured area above heart level to reduce swelling.
5. **Seek medical attention.** Fractures need imaging. Your bathroom does not have an X-ray machine (and if it does, we have different questions).

FAST Stroke Assessment — Formalized

The FAST assessment is a rapid screening tool for stroke. Time is brain — every minute of untreated stroke destroys approximately 1.9 million neurons.[33]

Component	Test	Positive sign
F — Face	Ask the person to smile	One side droops (facial palsy)
A — Arms	Ask the person to raise both arms	One arm drifts downward
S — Speech	Ask the person to repeat a simple phrase	Speech slurred or incomprehensible
T — Time	Note the time symptoms started	Call 112 immediately. Note onset time — it determines treatment eligibility.

Key detail: Thrombolytic therapy (clot-busting medication) is most effective within 4.5 hours of symptom onset. The treatment window is:

$$t_{treatment} < 4.5\text{ hours from onset}$$

If you don’t know when symptoms started, the clock starts at the last time you know the person was normal. This is why “Time” is in the acronym — it’s not just urgency, it’s a literal countdown.

Vital Signs (How to Check If You’re Okay-ish)

Sign	How to check	Normal range	Concerning	Critical
Heart rate (HR)	Two fingers on wrist (radial artery) or neck (carotid). Count beats for 15 seconds × 4.	$HR \in [60, 100]$ bpm	< 50 or > 120 at rest	< 40 or > 150
Respiratory rate (RR)	Count breaths for 30 seconds × 2. One inhale + exhale = 1.	$RR \in [12, 20]$/min	< 10 or > 30/min	< 8 or > 40/min
SpO_2 (oxygen saturation)	Pulse oximeter (if available).	$SpO_2 \in [95, 100]$%	90–94%	< 90%
Blood pressure (BP)	BP cuff (if available).	$BP < 120/80$ mmHg	120–139/80–89	$\geq 140/90$ or $< 90/60$
Consciousness	Can you answer: who are you, where are you, what day is it?	All 3 correct	Any incorrect or confused	Unresponsive
Skin color	Look at your face, lips, nail beds.	Pink/warm	Pale/blue/gray/mottled	Cyanotic (blue lips/nail beds)

If any vital sign is in the “concerning” column and you’re alone → call 112. If in the “critical” column → call 112 immediately, no deliberation. These numbers aren’t suggestions — they’re thresholds derived from decades of clinical data.

NRS Pain Scale Quick Reference

For a detailed breakdown of the NRS pain scale with physiological correlates, see Ch.3 (Situation C). Quick version:

$$NRS \in [0, 10]$$

- NRS 0–3: Manage with self-care
- NRS 4–6: Assess cause, consider professional help
- NRS 7–10: Call 112 if acute onset; 📍 Self Ambulance protocols

Nur

Mild

0	1	2
None	Barely noticeable	Minor
Baseline	Slight HR ↑	HR ↑ 10%

Pain NRS Correlates — pain scale with physiological markers

When to Call for Help

Call 112 immediately if:

- Uncontrolled bleeding that doesn't slow with 10 minutes of pressure
- Difficulty breathing or chest pain with jaw/arm radiation
- Loss of consciousness (even brief)
- Suspected stroke: face drooping, arm weakness, speech difficulty (remember: **FAST** — Face, Arms, Speech, Time)
- Severe allergic reaction: swelling of throat/tongue, difficulty breathing, widespread hives
- Seizure lasting > 5 minutes or first-time seizure
- Head injury with vomiting, confusion, or loss of consciousness
- Severe abdominal pain with rigidity (stomach hard as a board)
- Any situation where you think "I should probably call" — call

The golden rule: If you're debating whether to call, the debate itself is a signal. Call. Dispatcher > regret.

Self-Care While Waiting

1. **Position yourself safely.** Conscious and breathing fine? Sit or lie in a comfortable position. Semi-recumbent (propped up at 45°) works well. Feeling faint? Lie flat with legs elevated. Nauseous? Lie on your side (recovery position). The bathroom floor is a perfectly acceptable place for this.
2. **Stay conscious.** If you feel yourself fading, fight it — not heroically, just practically: talk to yourself out loud, keep your eyes open and focused on a specific object, move your fingers and toes. Sensory input keeps the brain online.
3. **Breathe.** → 📍 Calm Guide (Ch.4, Breathing Exercises). The same breathing techniques that help anxiety also help pain and shock.
4. **Don't eat or drink.** If you might need surgery, an empty stomach is safer. Small sips of water are usually fine, but ask the dispatcher.
5. **Unlock the door.** If you've called 112, emergency services need to be able to reach you. Unlock the bathroom door now, while you still can.
6. **Phone placement.** Put your phone on speaker or place it within arm's reach. Don't hold it in your hand if you might lose consciousness — it'll fall and become unreachable.

Self Ambulance for Non-Physical Emergencies

The same framework applies to situations where the "injury" is psychological or situational. The logic is identical: stabilize, assess, get help.

Physical version	Psychological equivalent
Uncontrolled bleeding	Acute panic / emotional flooding

Physical version	Psychological equivalent
Apply pressure	📍 Calm Guide breathing exercises
Elevate the wound	Remove yourself from the triggering situation
Call 112	Call a crisis line (→ Ch.7, §14.2)
Stay conscious	Stay present — **5-4-3-2-1 grounding**: name 5 things you can see, 4 you can touch, 3 you can hear, 2 you can smell, 1 you can taste
Unlock the door	Tell someone you need help
Monitor vital signs	Monitor your emotional state: "Am I getting worse, stable, or better?"

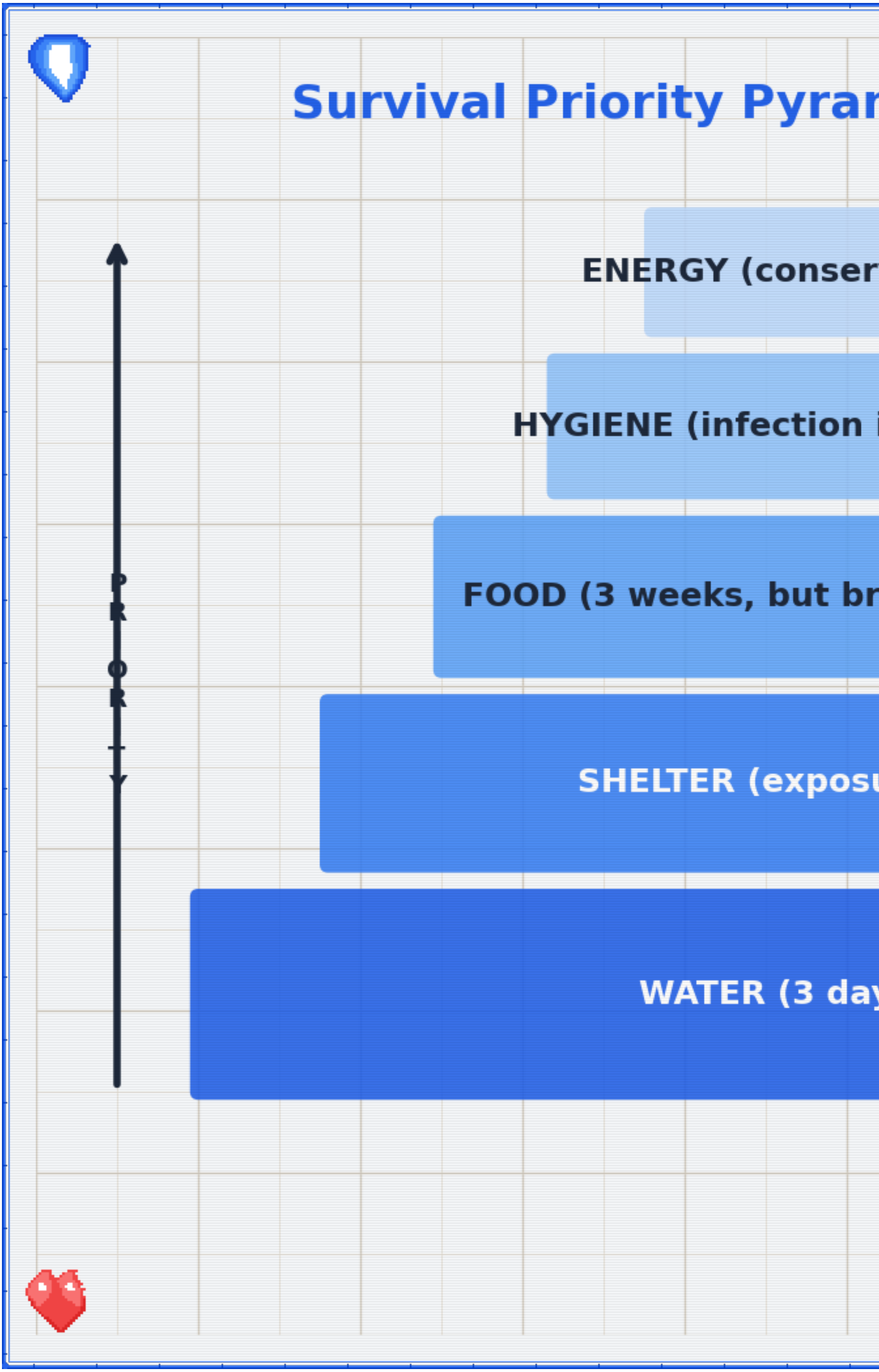
The 5-4-3-2-1 technique is worth memorizing because it requires zero equipment and works everywhere. It forces your brain to shift from internal distress to external perception. You can't process a threat response and catalog sensory data simultaneously — one displaces the other. Use that.

title: "Zombie Guide" chapter: 6 revision: "3.2.0" last_updated: "2026-05-03" dependencies: - ../diagrams/survival_pyramid.png - ../diagrams/scaling_chart.png - ../diagrams/survival_probability_function.png - ../diagrams/group_complexity_scaling.png - ../diagrams/water_requirements_scaling.png —

🧟 Zombie Guide — When "Calm" Isn't the Problem, "Alive" Is

You arrived here from one of two paths: either you created something that hatched during a post-apocalypse, or you followed a biological-entity path that led to zombification. Either way, the social contract you've been operating under has been revoked. The rules are different now. Let's learn the new ones.

> **Important framing:** This guide is half genuine survival knowledge and half thought experiment — because the skills for surviving a zombie apocalypse overlap with the skills for surviving any catastrophe: earthquake, flood, economic collapse, prolonged infrastructure failure. If you prepare for zombies, you're prepared for reality. That's the joke that isn't entirely a joke.


Survival Priority Pyra
ENERGY (conser
HYGIENE (infection
FOOD (3 weeks, but br
SHELTER (exposu
WATER (3 day
PRIORITY

Survival Probability Function

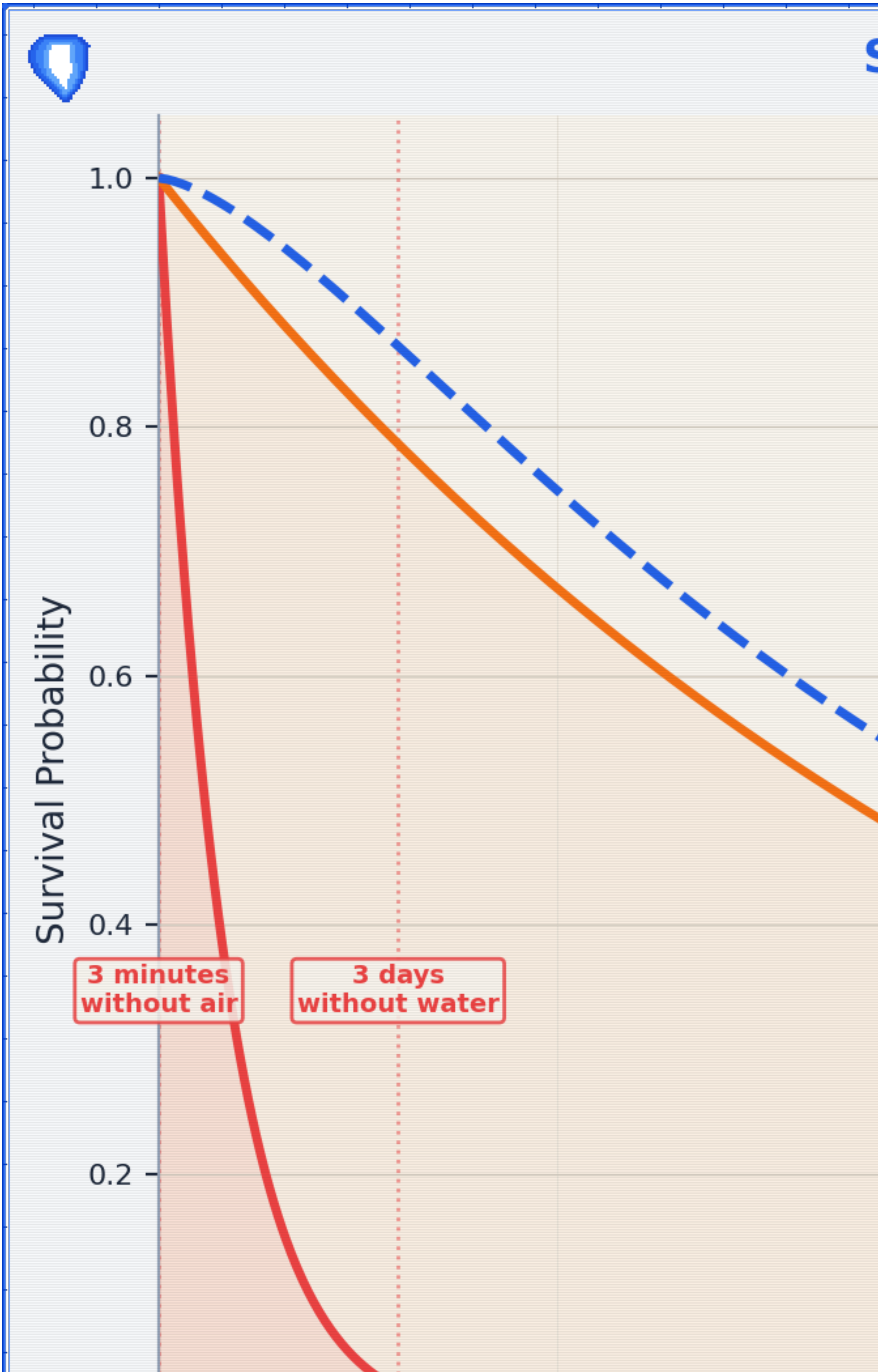
Before we dive into the practical stuff, let’s establish the mathematical framework. Your survival probability over time follows an exponential decay model — not because you’re doomed, but because risk accumulates:

$$P(\text{survive}, t) = e^{-\lambda t}$$

Where λ is the hazard rate (probability of a fatal event per unit time). The good news: λ is not fixed. Every skill you acquire, every resource you secure, every ally you find reduces λ. The survival function becomes:

$$P(\text{survive}, t) = e^{-(\lambda_0 - \sum_{i} r_i) t}$$

Where λ_0 is the baseline hazard and each r_i is a risk reduction factor: water secured, shelter established, group formed, perimeter defended. The math says: every preparation matters. Every skill moves the curve. You are not helpless — you’re a variable in your own survival equation.[34]



S
1.0
0.8
0.6
0.4
0.2
Survival Probability
3 minutes
without air
3 days
without water

Survival Probability Function — exponential decay with risk reduction factors

Mini Survival Guide (“In Nature”)

The infrastructure you’ve depended on is gone. The grid is down. You are, for the first time in your life, an animal in an ecosystem. Act like one — a smart one.

Water — Priority Zero

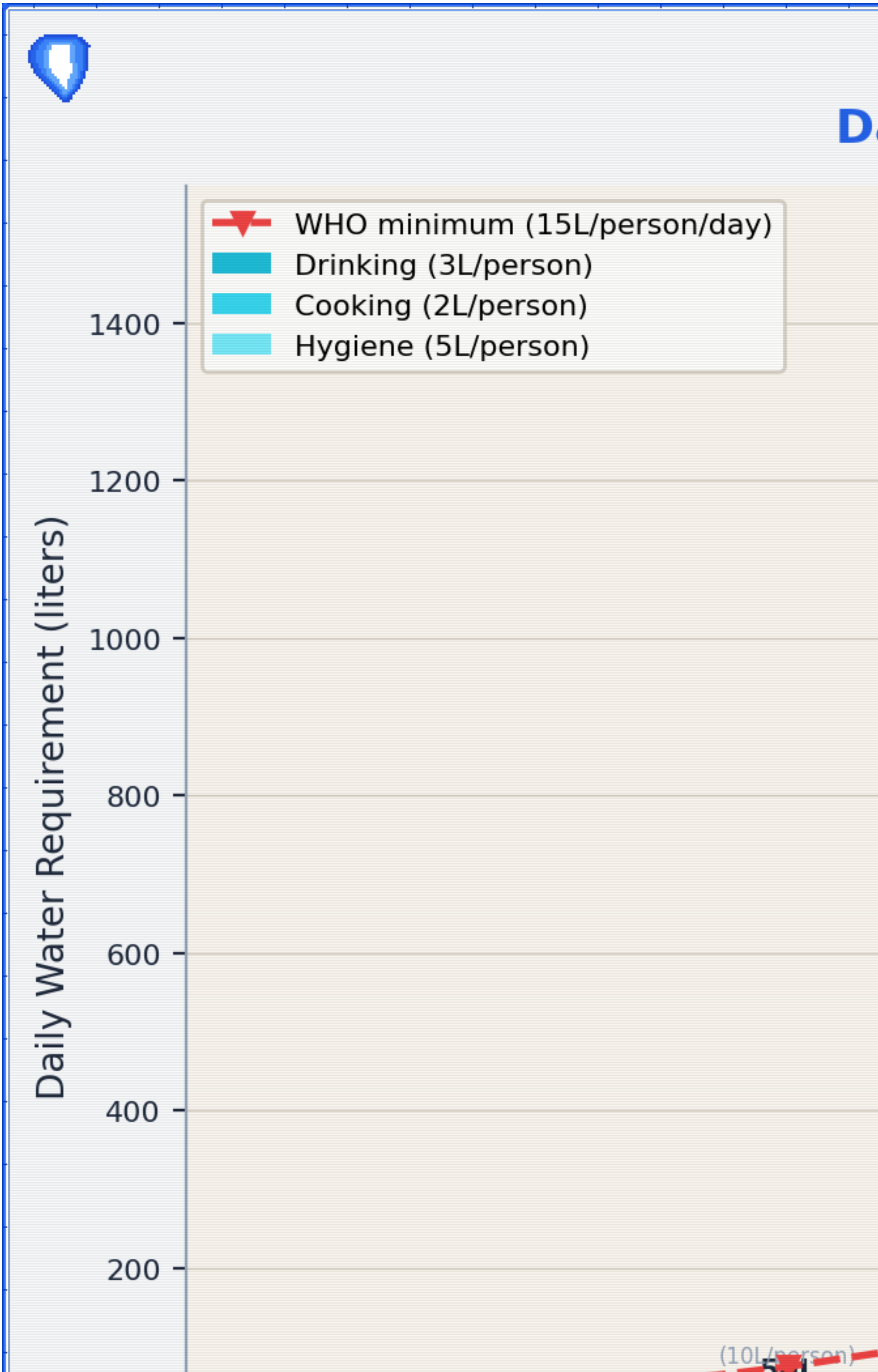
You die in 3 days without water. Everything else is secondary.

Water requirement formula:

$$W = n \cdot 3\text{L/day}$$

Where n is the number of people in your group and 3L/day covers drinking + basic cooking. This scales linearly but the logistics don’t:

Group size	Daily water need	Weekly water need	Collection effort
1	3L	21L	One person can carry
5	15L	105L	Multiple trips or coordinated
10	30L	210L	Dedicated water duty
30	90L	630L	Well or large water source required
100	300L	2100L	Infrastructure (well, pump, pipeline) needed


WHO minimum (15L/person/day)
Drinking (3L/person)
Cooking (2L/person)
Hygiene (5L/person)
Daily Water Requirement (liters)
1400
1200
1000
800
600
400
200

Water Requirements Scaling — daily water needs by group size

1. **Find it.** Flowing water (streams, rivers) is better than standing water (ponds, puddles). Rainwater is generally safe to collect. Dew can be gathered with cloth wrung into a container.
2. **Purify it.** Always assume water is contaminated. Methods, ranked by reliability:
    - **Boil** for 1 minute (3 minutes above 2000m elevation). Gold standard. Kill everything.
    - **Filter** through cloth + sand + charcoal layers (improvised). Removes particulates and some pathogens.
    - **Chemical treatment:** 2 drops of unscented household bleach per liter, wait 30 minutes. Or water purification tablets.
    - **UV exposure:** Clear plastic bottle, leave in direct sunlight for 6+ hours. UV kills pathogens. Slow but works.
3. **Store it.** Every container you find is now a water container. Fill everything. You will never have too much water.

Food

You can survive ~3 weeks without food. That's generous compared to water, but your cognitive function degrades fast without calories.

Caloric requirements:

$$E_{basal} \approx 1500 \text{ kcal/day} \quad \text{(resting metabolic rate)}$$ $$E_{survival} \approx 2500\text{–}3000 \text{ kcal/day} \quad \text{(active survival activity)}$$

The gap between basal and active survival requirements ($\Delta E \approx 1000\text{–}1500$ kcal) is the energy you spend on foraging, building, defending, and moving. Every unnecessary activity burns calories you can't afford to waste.

1. **Foraging.** Learn 3–5 easily identifiable local wild plants that are edible. Dandelion (entire plant), nettles (cook first), blackberries, wild garlic — widespread across Europe. If you can't identify it with certainty, don't eat it. Starvation is preferable to poisoning.
2. **Hunting/trapping.** Small game (rabbits, birds) is more practical than large game. Snares require wire and knowledge. Fishing requires water and patience. Principle: calories-in > calories-out. Don't spend more energy catching food than the food provides.
3. **Scavenging.** In a collapse scenario, preserved food in abandoned stores is your friend. Canned goods last years. Check seals — bulging cans are bacterial, not food. Prioritize: canned > dried > fresh (perishable).
4. **Insects.** Yes, really. Crickets, grubs, ants — high protein, widely available, no hunting skill required. Cook them if you can. If you can't, they're still more nutritious than your pride.

Shelter

Exposure kills faster than starvation. Hypothermia: wet + cold + wind.

1. **Location.** High ground (avoid flood zones), away from obvious paths (avoid detection), near water source but not directly beside it.
2. **Insulation.** Layers of dry leaves, pine needles, cardboard, clothing — anything between you and the ground. The ground is a heat sink.
3. **Windbreak.** A wall, hedge, car, or stacked debris blocking the prevailing wind direction. Wind strips heat. Block it.

Reading Traces & Observing Environments

- **Tracks.** Disturbed dirt, broken twigs, bent grass — direction and recency. Fresh tracks have sharp edges; old ones are rounded by wind and rain.
- **Sound.** Learn the baseline soundscape. Anything that breaks the baseline is information. Birds alarm-calling = predator nearby.
- **Smell.** Smoke = fire = other humans (or uncontrolled wildfire). Rotting = biological hazard. Cooking food = people, potentially friendly.
- **Patterns.** Zombies (canonical model) are attracted to sound and movement. Move quietly, deliberately, in shadow when possible.

Energy

Conserve it. The survival economy is brutally simple:

```
Energy in (food) ──→ Energy out (activity)

Rule: If the activity doesn't contribute to survival,
      don't do it. Panic running is energy waste.
      Strategic walking is energy investment.
```

Hygiene

The thing that kills most people in disaster scenarios isn’t the disaster itself — it’s the infection that follows. Cholera, dysentery, and wound sepsis have killed more humans than every zombie movie combined.

- **Defecate** away from water sources and camp. Bury it. Minimum 30m from water, 30m from shelter.
- **Wash hands** with ash and water if soap is unavailable. Ash is alkaline and breaks down oils/bacteria.
- **Wounds** must be cleaned immediately. Even small cuts can become lethal in a low-hygiene environment.
- **Teeth.** If you find a toothbrush, it’s more valuable than you think. Improvise: chewed stick (miswak technique), salt water rinse.

Mini Collapsing Society Guide

The immediate danger has passed (or you've adapted). Now: the infrastructure that kept 8 billion humans alive is degraded or gone. You need to rebuild some of it, locally.

Gathering Remaining Resources

Priority order — not by what's valuable, but by what's irreplaceable:

Priority	Resource	Why it matters	Notes
1	**Medical**	Drugs, equipment, and most critically: **medical personnel**	A trained nurse is worth 1000 crates of supplies. Protect them.
2	**Bicycles**	Transportation that doesn't require fuel. Silent. Maintainable.	Cars need fuel, fuel degrades, fuel is loud. Bikes are the post-apocalypse vehicle.
3	**Water source**	A well, spring, or clean river access point	Secure it. Defend it. Share it strategically.
4	**Energy source**	Solar panels, batteries, generators, firewood	Renewable > consumable. Solar panels are permanent; fuel is finite.
5	**Sustainable materials**	Seeds, soil, tools, fabric, rope, metal, knowledge	The stuff you can't loot — you have to produce.

Priority	Resource	Why it matters	Notes
6	**Tech**	Radios, phones (offline tools), reference books	A first-aid manual > smartphone with no network. Books don't need charging.

Looting etiquette (yes, even now): If someone else is already there, the resource is theirs by occupancy. Fight only if the cost of fighting is less than the value of the resource. It almost never is. Cooperation > conflict. Always.

Securing Friends + People in Need

You cannot survive alone long-term. The lone wolf dies. The pack lives.

1. **Find your people.** Start with people you know and trust. Expand outward. Trust is transitive but degrades with distance — one degree of separation is reliable, two is uncertain, three is stranger.
2. **Rescue those who need it.** Children, elderly, injured, isolated — these people cost resources short-term and provide social cohesion and meaning long-term. A community that abandons its vulnerable will collapse from the inside.
3. **Vetting newcomers.** Everyone gets a trial period. Observe behavior under stress. Do they share? Do they lie? Do they pull their weight? Three strikes is generous; two is practical.

Group Communication Channels

As your group grows, communication complexity explodes. The number of pairwise communication channels in a group of n people is:

$$C(n) = \frac{n(n-1)}{2}$$

Group size	Channels $C(n)$	Implication
3	3	Everyone talks to everyone. Easy.
5	10	Still manageable, but some conversations get missed.
10	45	You need structured communication.
20	190	Without systems, information is lost constantly.
50	1,225	Formal communication channels are mandatory.
100	4,950	Hierarchy or network structure required.

This is why small groups feel “tight” and large groups feel “bureaucratic” — the communication channels scale quadratically. You can’t fight the math. You can only design systems that acknowledge it.[35]

Dunbar Numbers — The Social Brain Hypothesis

Robin Dunbar’s research on primate neocortex size and social group sizes produced a series of concentric circles of human social capacity:[36]

$$D_n = 5 \cdot 3^{n-1} \quad \text{for } n \in \{1, 2, 3, 4, 5\}$$

Layer	Size	Relationship quality	In survival context
D_1	~5	Intimate support group	Your inner circle. Trust with your life.
D_2	~15	Close friends	Reliable allies. Regular contact.
D_3	~50	Casual friends	Know their names and skills. Occasional contact.
D_4	~150	Meaningful contacts	Recognize faces. Know reputations.
D_5	~500	Acquaintances	Names might ring a bell. Expanding network.

The implication for survival groups: at ~5 people, you have deep mutual knowledge. At ~15, you need explicit coordination. At ~50, you need sub-groups and representatives. At ~150, you need formal governance. The Dunbar numbers aren’t just social science — they’re the scaling limits of trust, and trust is the currency of survival.

Self-Defense (Last Resort)

The math of violence is always bad: winner is still injured and spent resources; loser is dead or vengeful; avoider is intact with resources preserved.

1. **Situational awareness.** See threats before they become fights. The best fight is the one that never happens.
2. **Barriers.** Doors, walls, fences, distance. Obstacles buy time, and time buys options.
3. **Improvised weapons.** Reach (sticks, pipes) > edge (knives, glass) > weight (rocks, hammers). Reach keeps them away. That’s the whole point.
4. **Numbers.** A group of 3 is harder to attack than a group of 1. This is the fundamental math of defense.

Water · Energy · Hygiene at Scale

- **Water:** 3 liters per person per day (drinking + cooking). 10 people = 30L/day. A well producing 100L/day supports ~30 people.

- **Energy:** One solar panel (~300W) supports basic lighting + phone charging for a small group. Heating and cooking require significantly more. Prioritize: cooking > light > comfort.
- **Hygiene:** Latrines must scale with population. 1 latrine per 20 people, minimum. A cholera outbreak in a group of 50 with no sanitation will kill more than any zombie ever did.

Mini Building Society Guide

You've survived. Now you need to live. Organizing, making decisions together, building something that outlasts the crisis.

Organizing: The First Meeting

1. **Agenda for Meeting #1:**
    - Who's here? (Names, skills, needs)
    - What do we have? (Resource inventory)
    - What do we need? (Gap analysis)
    - Who does what? (Role assignment)
    - When do we meet again? (Daily at first, then weekly)
2. **Roles to fill immediately:**
    - **Coordinator** — keeps the schedule, calls meetings, tracks decisions (facilitator, not leader)
    - **Supply manager** — tracks resources in and out
    - **Medic** — whoever has the most medical knowledge
    - **Security** — awareness, not aggression
    - **Cook** — communal meals build trust faster than any meeting

Ostrom's 8 Principles for Common-Pool Resource Governance

Elinor Ostrom won the Nobel Prize in Economics (2009) for her work on how communities govern shared resources without top-down control. Her 8 design principles are the closest thing to a proven blueprint for self-governance:[37]

1. **Clearly defined boundaries** — Who can use the resource? Who can't?
2. **Congruence with local conditions** — Rules fit the actual situation, not an abstract ideal
3. **Participatory decision-making** — People affected by the rules help make them
4. **Monitoring** — Someone watches the resource and the rule-followers
5. **Graduated sanctions** — Small violations get small penalties, not exile
6. **Conflict resolution mechanisms** — Fast, cheap, fair dispute resolution
7. **Minimal recognition of rights** — External authorities don't undermine local governance
8. **Nested enterprises** — Large systems are built from smaller, self-governing units

These principles emerged from studying fisheries, irrigation systems, forests, and pastures — real communities managing real scarce resources over centuries. They work because they align incentives with consequences. A community that follows all 8 is remarkably resilient. A community that ignores them is, historically, short-lived.

Forms of Self-Administration

Model	How it works	Best for	Weakness
Consensus	Everyone must agree. Discussion until unanimous.	Small groups (≤8)	Slow. One holdout blocks everything.
Consent	Decisions proceed unless someone objects with reason.	Small-medium (≤25)	Faster. Requires trust.
Voting	Majority wins. Simple, familiar.	Medium (≤100)	Minority gets overridden.
Delegation	Groups elect representatives who decide.	Large (100+)	Distance between decision-makers and affected people.
Rotation	Roles rotate on a fixed schedule.	Any size	Inconsistent leadership.

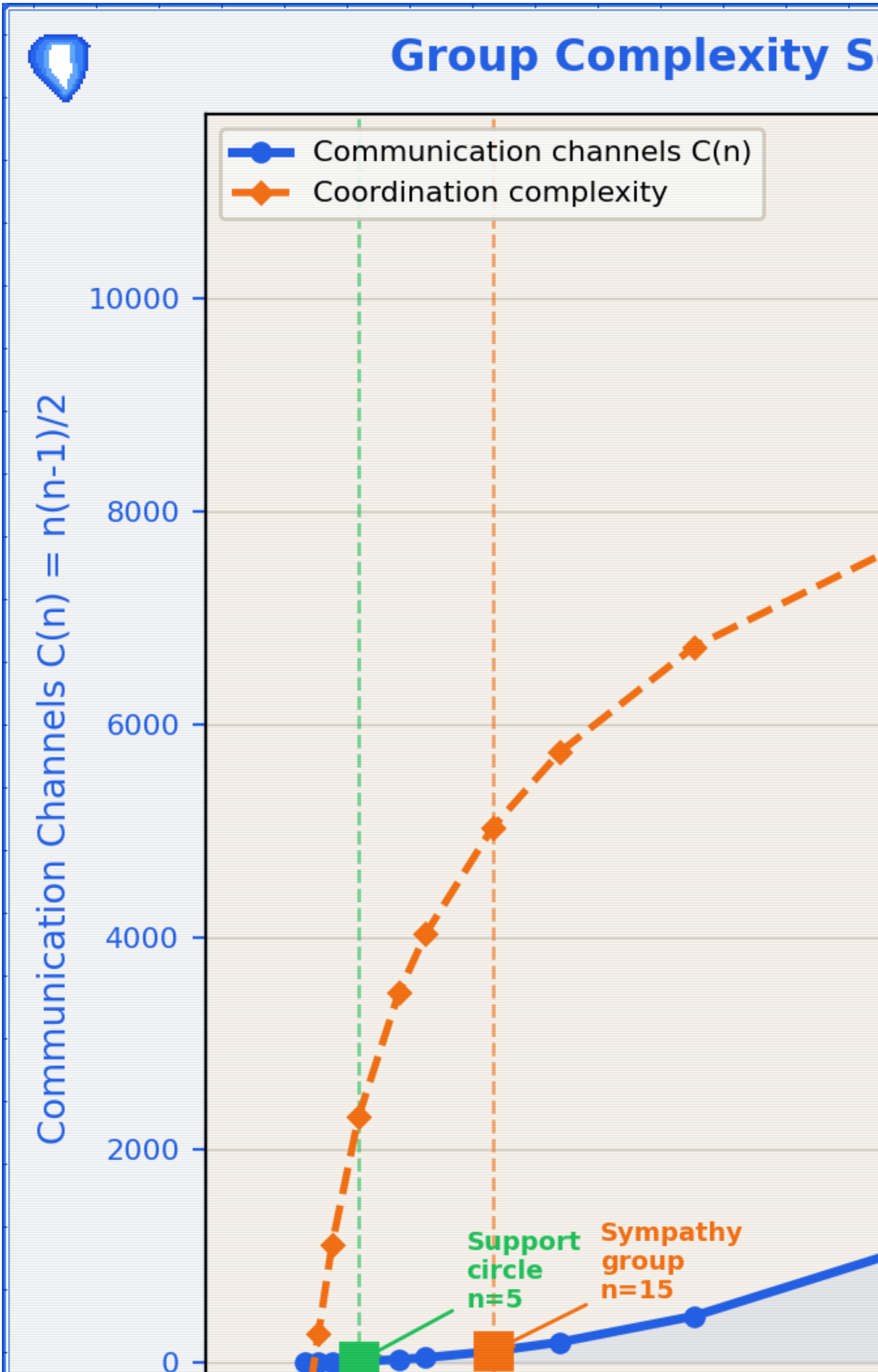
Recommendation: Start with Consent for day-to-day, Voting for big decisions. Adjust as you scale.

Psychology of the Masses / Sociology Basics

- **Us vs. Them.** The most powerful social force. It can unite against an external threat — or split from within. If someone is trying to convince you that a subgroup within your community is the enemy, that person is the actual threat.
- **Conformity pressure.** Solomon Asch showed people will deny the evidence of their own eyes to agree with a group. If everyone says the water is safe and you think it isn't — speak up. Dissent is a survival skill.
- **Diffusion of responsibility.** "Someone else will handle it." Assign specific tasks to specific people. "Someone check the perimeter" = nobody checks. "Anna, check the perimeter at 2100" = perimeter gets checked.
- **Authority capture.** In crisis, people defer to whoever seems most confident. Confidence ≠ competence. The quiet person with the notebook is probably right.

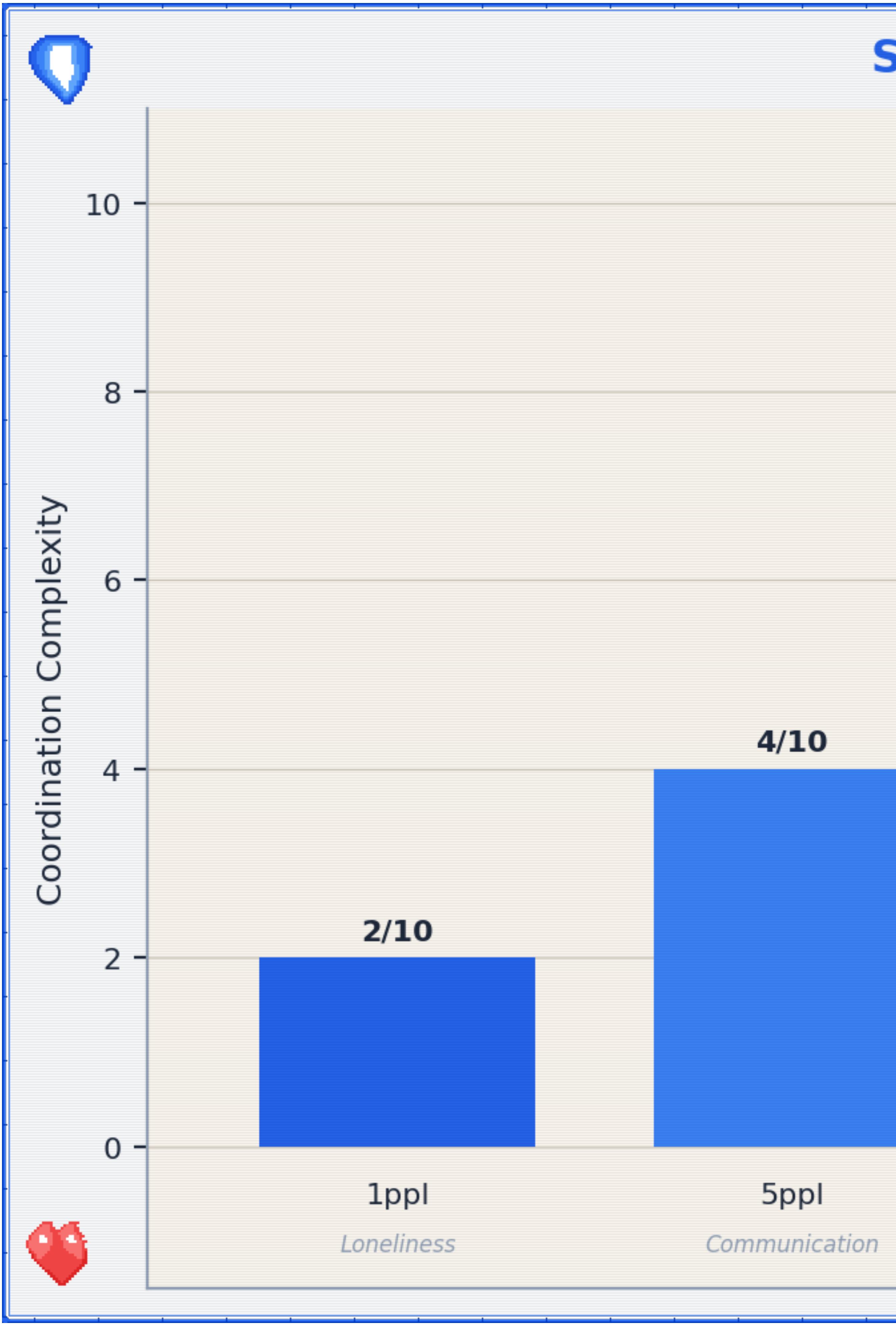
Group Complexity Scaling

As groups grow, the internal complexity grows faster than the headcount. We can model this as:


Group Complexity S
Communication channels C(n)
Coordination complexity
Communication Channels C(n) = n(n-1)/2
10000
8000
6000
4000
2000
0
Support
circle
n=5
Sympathy
group
n=15

Group Complexity Scaling — communication channels, governance layers, and management overhead

Scaling Requirements: 1 → 10 → 100



Coordination Complexity
10
8
6
4
2
0
2/10
4/10
1ppl
5ppl
Loneliness
Communication

Scaling Chart

Scale	Key challenge	What breaks	What to add	Dunbar layer
1 person	Loneliness, skill gaps	Nothing — it’s just you	Find people. Any people.	D_1 (you)
2–5	Communication, trust	Unspoken expectations	Explicit agreements, shared meals	D_1 (intimates)
6–10	Coordination, role clarity	“I thought you were doing that”	Regular meetings, written task lists	D_2 (close friends)
11–25	Decision speed	Consensus too slow	Switch to Consent or Voting	D_2→D_3 transition
26–50	Sub-groups form	Information silos	Communication channels, representatives	D_3 (casual friends)
51–100	Governance complexity	Power concentration	Formal roles, accountability, rotation	D_4 (meaningful contacts)
100+	Institutional memory	Forgetting why decisions were made	Written records, onboarding for newcomers	D_5 (acquaintances)

The pattern: every time you roughly double the group, your coordination systems need to level up. Don’t wait for things to break — anticipate the next scale and prepare before you reach it. The Dunbar numbers tell you approximately where the breakpoints are; Ostrom’s principles tell you how to build the structures that bridge them.

title: “Professional Support Directory” chapter: 7 revision: “3.2.0” last_updated: “2026-05-03” dependencies: [] —

📞 Professional Support Directory — When the Bathroom Isn’t Enough

This guide is a first response tool, not a replacement for professional help. The bathroom is where you stabilize. The people and organizations listed here are where you resolve. Every entry in this directory was referenced by at least one path in the main guide — this section is where those references land.

How to use this directory: Find the category that matches your need. Call the number, visit the website, or go to the place. If the first contact doesn't work, try the next one. Persistence is not weakness — it's logistics.

Regional note: Listings are primarily oriented toward Germany (D) with EU context. If you're reading this elsewhere, the categories still apply — substitute your local equivalents.

Emergency Numbers

Service	Number	When to call	Notes
Fire / Medical Emergency	**112**	Life-threatening medical emergencies, fires, accidents	EU-wide. Dispatcher stays on line. Speakerphone for hands free.
Police	**110**	Crimes in progress, threats, domestic violence	EU-wide. Can't speak? Dial, wait, press 55 (Germany).
Poison Control	030/19240 (Berlin)	Suspected poisoning, overdose, chemical exposure	Have substance name ready.
Crisis Line (general)	0800/111 0 111	Acute emotional crisis, suicidal thoughts	Free, 24/7, anonymous. No question too small.
Children & Youth Emergency	116 111	Children/teens in crisis	EU-wide. Also for adults concerned about a young person.

The 112/110 decision matrix:

```
◆ Is someone's life in immediate danger?

    ├─ Yes → 112 (ambulance + fire if needed)
    └─ No, but there's a crime/threat → 110 (police)
       No, and it's emotional/psychological → 0800/111 0 111
```

Psychological Support

Your brain is an organ. Sometimes organs need professional attention. This isn't character failure — it's maintenance.

Evidence-Based Therapy Effectiveness

When you're told to "get help," it helps to know that the help actually works. Here's what the evidence says about major therapeutic approaches:[38]

Therapy	Primary use	Response rate	Evidence level
CBT (Cognitive Behavioral Therapy)	Anxiety, depression, PTSD	~60% response rate for anxiety[39]	★★★ Multiple large RCTs and meta-analyses
EMDR (Eye Movement Desensitization)	PTSD, trauma	~70% PTSD symptom reduction[40]	★★★ WHO-recommended for PTSD
IPT (Interpersonal Therapy)	Depression, grief	~50–60% response rate for depression	★★★ Well-validated
Psychodynamic therapy	Personality, chronic patterns	~50% improvement; effects increase over time[41]	★★☆ Growing evidence base
MBSR (Mindfulness-Based Stress Reduction)	Stress, anxiety, chronic pain	~30–40% symptom reduction[42]	★★★ Kabat-Zinn's 8-week program, widely replicated

The response rates aren't 100% because human brains are not simple machines. But a 60% chance of significant improvement is substantially better than the 0% chance of improvement from sitting in a bathroom indefinitely. The "what if it doesn't work for me?" worry is understandable but statistically unwarranted — and if one approach doesn't work, another might.

IASC Intervention Pyramid

The Inter-Agency Standing Committee (IASC) defines a pyramid of mental health and psychosocial support in emergencies:[43]

```
                   /\
                  /  \
                 / 4. \      Specialized services
                / Clinical \   (psychiatry, specialized
therapy)
               /────────────────\
              /   3. Focused     \   Non-specialized support
             /  non-specialized   \  (trained counselors, support
groups)
            /──────────────────────\
           /  2. Community & family \   Social support
          /   support mechanisms     \  (group activities,
peer support)
         /────────────────────────────\
```

```
         /   1. Basic services & security    \   Foundational
needs
        /     (food, shelter, water, safety)   \  (this is where
the
       /──────────────────────────────────────────\  Zombie Guide
lives)
```

Key insight: Layer 1 must be secure before Layer 2 is meaningful, Layer 2 before Layer 3, and so on. This is why the guide prioritizes physical safety first — you can't do therapy when you're still in danger. The IASC pyramid is also why the Zombie Guide's survival section (Layer 1) comes before the Building Society section (Layer 2–3).

Crisis & Acute Support

Resource	Contact	What it offers	When to use
Telefonseelsorge	0800/111 0 111 (or 0 222)	Anonymous, free, 24/7 phone counseling	Acute emotional crisis, loneliness, despair
Nummer gegen Kummer	116 111 (youth) / 0800/111 0550 (parents)	Phone counseling for youth and parents	When a young person is struggling, or you're struggling as a parent
Psychiatrischer Krisendienst	Search: "[your city] psychiatrischer Krisendienst"	Mobile crisis team — they come to you	When you can't go to them. Many cities have 24/7 teams.

Ongoing Support

Resource	How to access	What it offers
Hausarzt (GP)	Make an appointment	First point of entry. Can refer to specialists, prescribe medication, sign you off work.
Psychotherapist	Search: therapie.de, psych-info.de, Jameda	Talk therapy, CBT, trauma processing. Waitlists can be long — ask about "Akutplätze."
Psychiatrist	Via GP referral or direct	Medication management, diagnosis, severe mental health conditions.
EAP (Employee Assistance)	Via your employer (if available)	Free, confidential short-term counseling. Often underused.
Self-help groups	Search: selbsthilfegruppe.de	Peer support for specific issues. Shared experience is powerful medicine.

Friends’ Psych Support Guide

When someone you care about is struggling and you don’t know what to do:

1. **Listen before fixing.** Ask: “Do you want me to just listen, or do you want suggestions?” Respect the answer.
2. **Don’t minimize.** Avoid “it could be worse” or “just think positive.” Well-intentioned and deeply unhelpful.
3. **Stay connected.** Check in regularly, even with a simple “thinking of you.” Isolation amplifies everything.
4. **Know your limits.** You are a friend, not a therapist. If the situation exceeds your capacity, say so with love: “I care about you and I think you need more support than I can give. Can I help you find someone?”
5. **Take care of yourself.** Empathy is finite. Refill your own reserves → 📍 Calm Guide (Ch.4).

Legal Support

When the situation involves law, rights, or potential criminal liability, you need someone who speaks the language of the system.

Self-Defense Law (Germany — §32 StGB Notwehr)

- You may defend yourself against a **present, unlawful attack** against yourself or another person.[44]
- The defense must be **necessary** — no less harmful alternative available.
- The defense must be **proportionate** — you can’t use deadly force to protect property alone.
- No duty to retreat in Germany, but retreating is always the safer legal strategy.
- If you acted in self-defense and are unsure about the legal outcome → contact a lawyer **before** making any statements to police.

The four elements of §32 StGB, formalized:

Element	Legal requirement	Practical meaning
(1) Present attack	An attack is happening now, not hypothetical	You can’t preemptively defend against an imagined threat
(2) Unlawful	The attack violates law	You can’t claim self-defense against lawful actions (e.g., police arrest)
(3) Defensive action	Your response is directed at stopping the attack	Vengeance after the fact ≠ self-defense
(4) Proportionality	The response is commensurate with the threat	No shooting someone for pushing you[45]

Finding Legal Help

Resource	Contact	What it offers
Notdienst (emergency lawyer)	0180/522 44 66	24/7 emergency legal advice
Rechtsanwaltskammer	Search: “[your city] Rechtsanwaltskammer”	Lawyer directory, free initial consultation referrals
Beratungshilfe	Local Amtsgericht	Free legal advice for low-income individuals
Fachanwalt für Strafrecht	Search: “[your city] Fachanwalt Strafrecht”	Criminal law specialist
Weisser Ring	0180/34 34 34	Support for crime victims — legal, financial, emotional

Medical Support

Reproductive Health

Resource	Contact	What it offers
Pro Familia	profamilia.de	Counseling pregnancy, contracepti abortion, sexual healt Non-judgmental Confidentia
Schwangerschaftskonfliktberatung	Search: “[your city] Schwangerschaftskonfliktberatung”	Mandatory counseling before abortion in Germany. Free.
Gynecologist (Frauenarzt)	Search: Jameda, Doctolib	Reproductiv health, prenatal ca contracepti

Pediatric / Child Health

Resource	Contact	What it offers
Kinderarzt	Search: Jameda, Doctolib	General child health, vaccinations, development checks.

Resource	Contact	What it offers
Kinder- und Jugendpsychiatrie	Via GP referral or direct	Mental health support for children and adolescents.
Kinderschutzbund	kinderschutzbund.de	Child welfare, abuse prevention, family support.

General Medical

Resource	When to use	Notes
112 / Rettungsdienst	Life-threatening emergencies	Don’t drive yourself. Ambulance = treatment starts en route.
Kassenärztlicher Bereitschaftsdienst	116 117	After-hours non-emergency medical care. Evenings, weekends, holidays.
Hausarzt	Everything non-urgent	Your coordination point. Go here first for anything not 112-worthy.

Social Support

State Child Support (Germany)

Benefit	Who it’s for	How to apply
Kindergeld	Parents of children < 18 (or < 25 if in education)	Via Familienkasse. €250/child/month (2026).
Elterngeld	Parents on parental leave	Via Elterngeldstelle. Up to 65–67% of previous income for up to 14 months.
Unterhaltsvorschuss	Single parents not receiving child support	Via Jugendamt. State pays missing support, then collects from other parent.
Wohngeld	Low-income households	Via local Wohngeldbehörde. Housing cost subsidy.
Bürgergeld	Unemployed or low-income individuals	Via Jobcenter. Basic living support + job search assistance.

Local Initiatives & Community

Type	How to find them	What they offer
Volksküche / Soup kitchens	Search: "Volksküche [your city]"	Free or cheap meals, community
Repair Cafés	repaircafe.org	Free repair sessions, skill sharing, social connection
Community gardens	Search: "Urban Gardening [your city]"	Food production, community, therapy-by-dirt
Meetup groups	meetup.com	Interest-based social groups — board games, hiking, coding
Religious communities	Local churches, mosques, temples	Community, support, many are explicitly welcoming
Queer centers	Search: "LSVD [your city]"	LGBTQ+ specific support, social events, counseling
Flüchtlingsrat	Search: "Flüchtlingsrat [your state]"	Support for refugees and migrants — legal, social, practical

Quick Reference Card

Tear this out or photograph it. Keep it in your phone.

```
┌──────────────────────────────────────────┐
│         EMERGENCY QUICK REFERENCE        │
├──────────────────────────────────────────┤
│  112    Fire / Medical Emergency         │
│  110    Police                           │
│  116117 After-hours doctor (non-emerg.)  │
│  0800/111 0 111  Crisis line (24/7)      │
│  116111 Youth emergency                  │
├──────────────────────────────────────────┤
│  Your doctor: ________________________   │
│  Local hospital: _____________________   │
│  Therapist: __________________________   │
│  Emergency contact: __________________   │
│  WiFi password: _____________________    │
├──────────────────────────────────────────┤
│  → Full directory: Ch.7                  │
│  → Calm Guide: Ch.4                      │
│  → Self Ambulance: Ch.5                  │
│  → IASC pyramid: p.7.2                   │
└──────────────────────────────────────────┘
```

title: "Appendix — The Loose Ends" chapter: 8 revision: "3.2.0" last_updated: "2026-05-03" dependencies: - ../diagrams/decision_flow_graph.png —

Appendix — The Loose Ends

This section collects the things that don't fit neatly into the flow but are essential nonetheless: the master cross-reference, fillable fields, the symbol legend, formula index, and a notes page. Think of it as the bathroom drawer where the important miscellaneous stuff lives.

Master Cross-Reference: Where Everything Points

Every path in this guide converges on one (or more) of four destinations. This table shows every section and where it routes.

Source section	Routes to	Diagram references (v3.2)
A1 (Not yet geschlüpft)	📍 Calm Guide · 📍 Prof. Support (§14.4)	—
A2 (Geschlüpft, minutes ago)	📍 Zombie Guide · 📍 Self Ambulance · 📍 Calm Guide	—
A3 (Geschlüpft, days ago)	📍 Prof. Support (§14.4, §14.2) · 📍 Calm Guide	—
A4 (< 10 years ago)	📍 Prof. Support (§14.5) · 📍 Calm Guide	—
A5 (> 20 years ago)	📍 Calm Guide	—
A6 (Ambiguous)	📍 Calm Guide · 📍 Prof. Support (§14.4, §14.5)	—
A7 (Harmed/ended life)	📍 Prof. Support (§14.3, §14.2) · 📍 Self Ambulance · 📍 Calm Guide	—
A8 (Managing responsibility)	📍 Calm Guide · 📍 Prof. Support (§14.3, §14.2)	—
B1 (Acute anxiety)	📍 Calm Guide	`anxiety_severity_spectrum.png` `anxiety_severity_spectrum.png`

Source section	Routes to	Diagram references (v3.2)
B2 (General anxiety)	📍 Prof. Support (§14.2)	
B3 (Worried about...)	📍 Prof. Support (§14.2) · 📍 Calm Guide	—
C1 (Physical pain)	📍 Self Ambulance	pain_nrs_correlates.png
C2 (Psych pain)	📍 Calm Guide · 📍 Prof. Support (§14.2)	—
D (Endangered)	📍 Prof. Support (§14.3, §14.2) · 📍 Calm Guide	—
E (Congesting)	📍 Self Ambulance · 📍 Calm Guide · 📍 Prof. Support (§14.2)	—
F (Bad smell)	📍 Calm Guide (minor)	—
G (No place to go)	📍 Prof. Support (§14.2, §14.5) · 📍 Calm Guide	—
Ch.1 (How to Use)	All destinations	decision_flow_graph.png
Ch.4 (Calm Guide)	📍 Prof. Support (Ch.7)	stress_decay_curve.png
Ch.5 (Self Ambulance)	📍 Calm Guide · 112	triage_priority_heatmap.png, pain_nrs_correlates.png
Ch.6 (Zombie Guide)	📍 Self Ambulance · 📍 Calm Guide	survival_probability_function.png, group_complexity_scaling.png, water_requirements_scaling.png

Most-referenced destinations (by frequency):

1. 📍 Calm Guide — 13 references (heart of the document)
2. 📍 Professional Support Directory — 11 references
3. 📍 Self Ambulance Guide — 4 references
4. 📍 Zombie Guide — 1 reference (but what a reference)

Diagram Index (v3.2)

All diagrams referenced in the guide, organized by chapter:

Diagram	Chapter	Description
`master_flowchart.png`	Ch.1	Master flowchart — all 7 entry points → 4 destinations
`situation_a_tree.png`	Ch.2	Situation A decision tree
`breathing_techniques.png`	Ch.4	Box Breathing, 4-7-8, Physiological Sigh
`survival_pyramid.png`	Ch.6	Survival priority pyramid
`scaling_chart.png`	Ch.6	Scaling requirements 1→100
`triage_flow.png`	Ch.5	Self Ambulance triage flow
`decision_flow_graph.png`	Ch.1	Guide topology as directed graph
`stress_decay_curve.png`	Ch.4	Cortisol decay with/ without active calming
`anxiety_severity_spectrum.png`	Ch.3	GAD-7 scoring with recommended routing
`triage_priority_heatmap.png`	Ch.5	Severity × urgency triage matrix
`survival_probability_function.png`	Ch.6	Exponential decay with risk reduction factors
`group_complexity_scaling.png`	Ch.6	Communication channels and governance layers
`pain_nrs_correlates.png`	Ch.3, Ch.5	NRS pain scale with physiological markers
`water_requirements_scaling.png`	Ch.6	Daily water needs by group size

Formula Index

All mathematical formulas used in the guide, with location and plain-language summary:

Formula	Location	What it means
$G = (V, E)$	Ch.1	The guide is a directed graph with vertices and edges

Formula	**Location**	**What it means**
$GAD\text{-}7 \in [0, 21]$	Ch.3	Anxiety severity score ranges from 0 to 21
$NRS \in [0, 10]$	Ch.3, Ch.5	Pain scale ranges from 0 to 10
$S(t) = S_0 + A(1 - e^{-\alpha t})$	Ch.3	Stress response activation curve
$t_{reset} \approx 10\text{min}$	Ch.4	Sympathetic-parasympathetic transition time
$C(t) = C_0 \cdot e^{-\lambda t}$	Ch.4	Cortisol decay function
$\lambda = \frac{\ln 2}{t_{1/2}}$	Ch.4	Decay constant from half-life
$HRV = \frac{SD_{RR}}{\overline{RR}}$	Ch.4	Heart rate variability coefficient of variation
$\text{CO}_2\text{ offload} \propto \frac{A \cdot \Delta P}{d}$	Ch.4	Physiological sigh CO_2 clearance rate
$\text{Cognitive Load} = I + E + G$	Ch.3	Working memory: intrinsic + extraneous + germane
$HR \in [60, 100]$ bpm	Ch.5	Normal heart rate range
$RR \in [12, 20]$/min	Ch.5	Normal respiratory rate range
$SpO_2 \in [95, 100]$%	Ch.5	Normal oxygen saturation range
$BP < 120/80$ mmHg	Ch.5	Normal blood pressure threshold
$t_{treatment} < 4.5$h	Ch.5	Stroke thrombolysis treatment window
$P(\text{survive}, t) = e^{-\lambda t}$	Ch.6	Survival probability function
$W = n \cdot 3\text{L/day}$	Ch.6	Water requirements by group size
$C(n) = \frac{n(n-1)}{2}$	Ch.6	Group communication channels
$D_n = 5 \cdot 3^{n-1}$	Ch.6	Dunbar number layers
$E_{basal} \approx 1500$ kcal/day	Ch.6	Basal metabolic caloric requirement
	Ch.6	

Formula	Location	What it means
$E_{survival} \approx 2500\text{–}3000$ kcal/day		Active survival caloric requirement

Fillable Fields

These are the fields left blank throughout the guide for you to personalize. Fill them in once, photograph the page, and you'll always have the information.

Field	Your value
WiFi password	
Your doctor (Hausarzt)	
Your doctor's phone	
Local hospital	
Local hospital address	
Therapist (if applicable)	
Therapist's phone	
Emergency contact #1	
Emergency contact #1 phone	
Emergency contact #2	
Emergency contact #2 phone	
Nearest pharmacy (Apotheke)	
Pharmacy night service	
Preferred comfort item location	
Scent candle location	
Seneca / Tao Te Ching location	
Local-server access	
Bike lock combination	
Spare key location	

Flowchart Symbol Legend (Extended)

Symbol	Name	Meaning
→	Arrow / flow	Follow this path in the indicated direction
◆	Decision diamond	A question with two or more possible answers — choose one to proceed
📍	Destination marker	A shared guide chapter — go there for detailed instructions
§	Terminus	

Symbol	Name	Meaning
		This path ends. No useful guidance beyond this point. Reconsider.
🔁	Loop	Return to an earlier point in the flowchart and re-enter from a different path
✓	Checkpoint	Verify a condition before proceeding
⚠	Warning	Critical safety note — do not skip
💡	Tip	Optional but helpful advice
`$...$`	Inline math	A formula within text — skip if not needed, plain explanation follows
`$$...$$`	Display math	A standalone formula — same rule, skip-friendly
`[^n]`	Footnote	Additional source or context — read if curious, skip if not

Color Coding

Color	Represents	Used for
Math blue	Primary paths, main headings	Default reading flow
Pure white	Background, breathing room	Pages, decision spaces
Green	Safe / calm destinations	Calm Guide references
Red	Urgent / danger routes	Emergency calls, life-threat alerts
Orange	Caution / decision points	◆ Decision diamonds
Purple	Zombie / speculative paths	Zombie Guide references
Gray	Supplementary / optional	Tips, notes, fillable fields
Cyan	Scientific / mathematical	Formulas, data tables, research citations

The Complete Decision Tree (Text Version)

For when you can't read the visual flowchart (low light, visual impairment, or the diagram got coffee on it), here's the entire guide compressed into a single navigable text tree:

```
BATHROOM EMERGENCY GUIDE — MASTER TREE (v3.2)
═══════════════════════════════════════════════

1. WHAT BROUGHT YOU HERE?
   ├── A: CAUSED TROUBLE (involving a life/entity)
   │   ├── Step 1: Biological or Silicon?
   │   │   ├── Silicon → Escaped? → BSI / talk to host / → see
```

```
Bio
   │   │   └── Biological → Step 2
   │   ├── A1: Not geschlüpft → medical support + → Prof. Support
§14.4
   │   │   └── Gestation milestones: 4w/8w/12w/20w/24w/28w/32w/
37w+
   │   ├── A2: Geschlüpft, minutes ago
   │   │   ├── Post-apocalypse → → Zombie Guide
   │   │   └── Pre-apocalypse → 110 / find humans / → Self
Ambulance
   │   ├── A3: Geschlüpft, days ago → medical + ethical → Prof.
Support §14.4, §14.2
   │   ├── A4: <10 years → state support + development guide →
Prof. Support §14.5 + → Calm
   │   │   └── WHO milestones: sit ~6mo, walk ~12mo, vocab
~200w@24mo
   │   ├── A5: >20 years → grow up pls / → Calm
   │   ├── A6: Ambiguous
   │   │   ├── (a) Don't want to but will → → Calm + → Prof.
Support
   │   │   └── (b) Want to but can't → money/medical/reproduction
→ Prof. Support §14.4, §14.5
   │   ├── A7: Harmed/ended life
   │   │   ├── Self-defense → §32 StGB (4 elements) + → Prof.
Support §14.3 + → Calm
   │   │   ├── Accident → → Self Ambulance + find humans + → Calm
   │   │   └── For fun/profit → §
   │   └── A8: Managing responsibility → → Calm + → Prof. Support
§14.3, §14.2
   │
   ├── B: FEEL ANXIOUS
   │   ├── B1: Acute → → Calm
   │   ├── B2: General → → Prof. Support §14.2
   │   ├── B3: Worried about… → friends/self/past/future → → Calm
+ → Prof. Support §14.2
   │   └── GAD-7: 0-4 minimal, 5-9 mild, 10-14 moderate, 15-21
severe
   │
   ├── C: FEEL PAIN
   │   ├── C1: Physical → → Self Ambulance
   │   ├── C2: Psychological → → Calm (+ → Prof. Support §14.2)
   │   └── NRS: 0-3 mild, 4-6 moderate, 7-9 severe, 10 emergency
   │
   ├── D: FEEL ENDANGERED
   │   ├── Imminent → 110/112 FIRST
   │   ├── Stabilized → → Prof. Support §14.3, §14.2 + → Calm
   │   └── Stress response: C(t) = C₀·e^(-λt), t½ ≈ 75 min
   │
   ├── E: THINGS CONGESTING → → Self Ambulance → → Calm → (→
Prof. Support §14.2)
   │   └── Cognitive load: 7±2 items (Miller), 4±1 chunks (Cowan)
   │
```

├── F: BAD SMELL → SO_2 (matches) → scent → ventilate → (→ Calm minor)
│
└── G: NO PLACE TO GO → hotlines → local initiatives → → Prof. Support §14.2, §14.5 + → Calm

DESTINATION GUIDES

📍 CALM GUIDE (Ch.4)
├── 4.1: You're allowed to be here (10-min Me Time; t_reset ≈ 10min)
├── 4.2: Breathing (Box / 4-7-8 / Physiological Sigh)
│   Stress decay: $C(t) = C_0 \cdot e^{-\lambda t}$
│   HRV = SD_RR / RR_mean
├── 4.3: Comfort inventory (dopamine + oxytocin mechanisms)
├── 4.4: Eventually: leaving the bathroom
└── 4.5: Seek help now or afterwards

📍 SELF AMBULANCE GUIDE (Ch.5)
├── 5.1: Life-threatening? → 112
│   Bayes' triage: P(critical|signs)
├── 5.2: First aid quick ref (wounds/burns/fractures/vitals)
│   Vitals: $HR \in [60,100]$, $RR \in [12,20]$, $SpO_2 \in [95,100]$, BP<120/80
│   FAST stroke: $t_{treatment} < 4.5h$
│   Burns: Lund-Browder chart for TBSA%
├── 5.3: When to call for help
├── 5.4: Self-care while waiting
└── 5.5: Non-physical emergencies (5-4-3-2-1 grounding)

📍 ZOMBIE GUIDE (Ch.6)
├── 6.1: Survival (water/food/shelter/traces/energy/hygiene)
│ $P(survive,t) = e^{-\lambda t}$, $W = n \cdot 3L/day$
│ $E_{basal} \approx 1500$ kcal, $E_{survival} \approx 2500\text{-}3000$ kcal
├── 6.2: Collapsing society (resources/friends/defense/scaling WASH)
│ $C(n) = n(n-1)/2$ communication channels
│   Dunbar: $D_n = 5 \cdot 3^{(n-1)}$ → 5/15/50/150/500
└── 6.3: Building society (organizing/governance/psychology/scaling 1→100)
          Ostrom's 8 principles for common-pool resource governance

📍 PROFESSIONAL SUPPORT DIRECTORY (Ch.7)
├── 7.1: Emergency numbers
├── 7.2: Psychological support
│   CBT ~60% response, EMDR ~70% PTSD reduction
│   IASC 4-layer intervention pyramid
├── 7.3: Legal support (§32 StGB: 4 elements)

```
    ├── 7.4: Medical support
    └── 7.5: Social support
```

Notes Page

Your space. Use it for whatever the flowcharts don't cover — local information, personal reminders, the name of that therapist your friend recommended, or just processing by writing.

title: "Version History" chapter: 9 revision: "3.2.0" last_updated: "2026-05-03" type: appendix dependencies: [] —

Version History

This appendix tracks all revisions of the Bathroom Emergency Guide. Per-chapter revision numbers are maintained in each file's YAML frontmatter; project-level history is recorded here and in `CHANGELOG.md`.

Revision Log

Version	Date	Author	Changes
3.2.0	2026-05-03	Bathroom Guide Project	**Scientific rigor update.** LaTeX math notation throughout (\(…\) / \[…\]). Pandoc footnotes with source citations added to all chapters. New scientific content: GAD-7 scoring (Ch.3), NRS pain scale with physiological correlates (Ch.3/5), cortisol decay function (Ch.4), HRV formula (Ch.4), physiological sigh CO_2 offload mechanism (Ch.4), Huberman Lab RCT citation (Ch.4), dopamine/ oxytocin comfort mechanisms (Ch.4), cognitive load theory — Miller's 7±2 and Cowan's 4±1 (Ch.3), SO_2 odor neutralization chemistry (Ch.3), stress response activation curve

Version	Date	Author	Changes
			(Ch.3), vital signs with formal ranges (Ch.5), FAST stroke formalization with treatment window (Ch.5), burn surface area — Lund-Browder chart (Ch.5), Bayes' triage probability (Ch.5), survival probability function (Ch.6), water requirement formula (Ch.6), group communication channel formula (Ch.6), Dunbar numbers (Ch.6), Ostrom's 8 principles (Ch.6), calorie requirements (Ch.6), evidence-based therapy effectiveness data (Ch.7), IASC intervention pyramid (Ch.7), §32 StGB four-element formalization (Ch.7), gestation milestone table with viability data (Ch.2), WHO child development milestones (Ch.2), caregiver burnout data (Ch.2). 8 new diagram references: decision_flow_graph, stress_decay_curve, anxiety_severity_spectrum, pain_nrs_correlates, triage_priority_heatmap, survival_probability_function, group_complexity_scaling, water_requirements_scaling. Formula index and diagram index added to Appendix (Ch.8).
3.0.0	2026-05-01	Bathroom Guide Project	Major content expansion: enriched all chapter bodies to 150+ words per section, added fillable fields, expanded professional support directory, added Sources & Further Reading chapter (Ch.10), refined decision trees.
2.0.0	2026-04-29	Bathroom Guide Project	Major visual and content update: pixel art sprites (12), enhanced diagrams with scanline/border/corner

Version	Date	Author	Changes
			effects, sleeker CSS with gradient headings, dark mode, card-style sections, cyan/magenta/teal/amber accent palette, dedicated HTML cover page for WeasyPrint, new Sources & Further Reading chapter (Ch.10), build pipeline v2.
1.0.0	2026-04-29	Bathroom Guide Project	Initial release. 10 chapters (00–09), 6 diagrams, 4 output formats, build pipeline.

Chapter Revisions

Chapter	File	Current revision	Last updated	Key v3.2 additions
0	00-cover.md	3.2.0	2026-05-03	Version bump, scientific rigor in description, updated doc control table
1	01-how-to-use.md	3.2.0	2026-05-03	Math notation legend, graph-theoretic model \(G=(V,E)\), decision_flow_graph diagram, footnotes
2	02-situation-a.md	3.2.0	2026-05-03	Gestation milestone table, WHO child development milestones, §32 StGB elements, cortisol/decision footnote, caregiver data
3	03-situations-b-g.md	3.2.0	2026-05-03	GAD-7 severity spectrum, NRS pain scale with physiological correlates, stress response curve \(S(t)\), cognitive load theory (Miller, Cowan), SO_2 neutralization chemistry, anxiety_severity_spectrum + pain_nrs_correlates diagrams
4	04-calm-guide.md	3.2.0	2026-05-03	Stress decay curve \(C(t)=C_0 e^{-\lambda t}\), 10-min rule formalization, HRV formula, physiological sigh CO_2 mechanism, Huberman Lab RCT, comfort inventory neurochemical

Chapter	File	Current revision	Last updated	Key v3.2 additions
				mechanisms (dopamine, oxytocin, olfactory, hydration, warmth, touch), stress_decay_curve diagram
5	05-self-ambulance.md	3.2.0	2026-05-03	Vital signs formal ranges (HR, RR, SpO_2, BP), FAST stroke formalization with $t<4.5$h window, Lund-Browder burn surface area, Bayes' triage probability, NRS pain correlation, triage_priority_heatmap + pain_nrs_correlates diagrams
6	06-zombie-guide.md	3.2.0	2026-05-03	Survival probability $P(t)=e^{-\lambda t}$, water formula $W=n\cdot 3L/day$, communication channels $C(n)=n(n-1)/2$, Dunbar numbers $D_n=5\cdot 3^{n-1}$, Ostrom's 8 principles, calorie requirements (E_{basal}, $E_{survival}$), survival_probability_function + group_complexity_scaling + water_requirements_scaling diagrams
7	07-professional-support.md	3.2.0	2026-05-03	Therapy effectiveness data (CBT ~60%, EMDR ~70%), IASC intervention pyramid, §32 StGB four-element table, evidence-based footnotes
8	08-appendix.md	3.2.0	2026-05-03	Diagram index (14 diagrams), formula index (21 formulas), updated cross-reference table with diagram refs, extended color coding (cyan = scientific), updated master tree with formulas
9	09-version-history.md	3.2.0	2026-05-03	Added v3.0.0 and v3.2.0 entries, updated chapter revision table with v3.2 additions column

Chapter	File	Current revision	Last updated	Key v3.2 additions
10	10-sources.md	3.2.0	2026-05-03	Added sections: Anxiety Assessment (GAD-7), Pain Assessment (NRS), Stress & Cortisol Research, Heart Rate Variability (HRV), Physiological Sigh / Breathing Research, Cognitive Load Theory, Dunbar Social Brain, Ostrom Commons Governance, IASC Guidelines, Therapy Effectiveness, Olfactory Science, Caregiver Burnout, Stroke Treatment, Burn Assessment

Build Metadata

Field	Value
Build system	Bash + Pandoc + WeasyPrint + matplotlib + Pillow
Source format	Markdown with YAML frontmatter + LaTeX math + Pandoc footnotes
Primary PDF route	Cover HTML + Guide HTML → WeasyPrint PDF
Fallback PDF route	Pandoc → LaTeX → PDF
Diagram format	PNG (200 DPI) with pixel art enhancements
Pixel art format	PNG (32×32 sprites, RGBA)
CSS version	2.0.0 (gradient headings, dark mode, pixel art classes)
Math rendering	Pandoc → HTML (MathJax/KaTeX), LaTeX (native), DOCX (OMML)

Update Protocol

1. Edit the relevant chapter file in `src/chapters/`.
2. Increment the `revision` field in that file's YAML frontmatter.
3. Add an entry to the Chapter Revisions table above.
4. Add a changelog entry to `CHANGELOG.md`.
5. Rebuild: `./bin/build_all.sh`
6. Verify outputs in `build/`.

title: "Sources & Further Reading" chapter: 10 revision: "3.2.0" last_updated: "2026-05-03" type: appendix dependencies: [] —

Sources & Further Reading

This guide didn't spring from a bathroom fully formed. The techniques, protocols, and recommendations throughout these pages are grounded in real, verifiable sources — from WHO guidelines and military survival manuals to Stoic philosophy and social psychology research. This section collects the most important references so you can go deeper on any topic that caught your attention (or your anxiety).

How to use this section: Each category corresponds to a major theme in the guide. Entries are marked with credibility tiers: ★★★ = official/primary source, ★★☆ = reputable secondary source, ★☆☆ = general web resource. When in doubt, trust the ★★★ sources first — they're the ones written by people with degrees and institutional backing, not bathroom philosophers (present company excepted).

First Aid & Emergency Response

The Self Ambulance Guide (Ch.5) draws on these sources for its triage protocols, wound care, burn treatment, and CPR guidance.

Source	URL	Notes
Red Cross — First Aid Steps	redcross.org/take-a-class/first-aid	★★★ Printable step-by-step first aid covering CPR, wound care, and emergency response
German Red Cross (DRK) — Erste Hilfe	drk.de/hilfe-in-deutschland/erste-hilfe	★★★ Official DRK first aid page with videos, course finder, and emergency procedures
IFRC — First Aid	ifrc.org/our-work/health-and-care/first-aid	★★★ Global Red Cross/Red Crescent first aid training overview across 191 national societies
ERC Guidelines 2025	erc.edu/science-research/guidelines	★★★ Latest European resuscitation guidelines — the European standard for CPR practice
St. John Ambulance — First Aid Reference Guide	sja.ca (PDF)	★★★ Comprehensive evidence-based first aid reference guide
St. John Ambulance NSW — Guides	stjohnnsw.com.au/first-aid-guides	★★☆ Up-to-date first aid advice from St. John Ambulance Australia

Survival & Emergency Preparedness

The Zombie Guide (Ch.6) is secretly a real survival guide. These sources cover the genuine skills that double as zombie preparation and actual catastrophe readiness.

Source	URL	Notes
SAS Survival Handbook	Wikipedia overview	★★☆ John "Lofty" Wiseman's classic survival manual — the book to have when civilization stops
WHO — Emergency Treatment of Drinking-Water	WHO Technical Note (PDF)	★★★ Field water treatment methods for emergencies
FEMA — Ready.gov	ready.gov	★★★ Official U.S. preparedness site: emergency plans, disaster kits, shelter, real-time alerts
FEMA — Basic Preparedness (PDF)	fema.gov (PDF)	★★★ Comprehensive FEMA guide: emergency plans, supply kits, shelter, community resources
BBK — Vorsorgen für Krisen und Katastrophen	bbk.bund.de	★★★ Official German Federal Office for Civil Protection emergency preparedness guide
BBK — Personal Preparedness (English)	bbk.bund.de/EN	★★★ English version of BBK's personal preparedness guidance
US Army Survival Manual FM 21-76	Archive.org (PDF)	★★★ Full PDF of the classic military survival manual — covers all climates and terrains

Electricity Safety & Basics

For when the lights go out (or you're wondering if that wire is safe to touch), these sources cover electrical fundamentals and emergency procedures.

Source	URL	Notes
OSHA — Electrical Safety Overview	osha.gov/electrical	★★★ Official OSHA page on electrical safety standards, shock, electrocution, and fire hazards

Source	URL	Notes
OSHA — Controlling Electrical Hazards (PDF)	OSHA Publication 3075	★★★ Comprehensive guide on identifying and controlling electrical hazards
HSE (UK) — Introduction to Electrical Safety	hse.gov.uk/ electricity	★★★ UK Health and Safety Executive guide to electrical safety at work and home
VDE — The Technology Organization	vde.com/en	★★★ German Association for Electrical, Electronic & Information Technologies — sets DIN VDE standards
VDE Standards Overview	vde.com/ standards-en	★★★ Overview of VDE's full spectrum of electrical engineering standards publications
ILO — Electrical Safety Guidelines	ilo.org	★★★ International Labour Organization electrical safety guidance for inspectors and practitioners

Nutrition & Emergency Food

The Zombie Guide's food section is grounded in these nutritional science sources. Whether you're foraging in the wild or rationing during a supply disruption, these references tell you what your body actually needs.

Source	URL	Notes
WHO — Nutrition Guidelines Portal	who.int/health-topics/nutrition	★★★ WHO's central portal for nutrition guidelines, policies, and global recommendations
WHO — Food and Nutrition Needs in Emergencies (PDF)	WHO Technical Guidance	★★★ Minimum energy, protein, fat, and micronutrient requirements in emergency food rations
DGE — German Nutrition Society	dge.de/english	★★★ Germany's authority on nutritional science — official English page
DGE — Food-Based Dietary Guidelines	dge.de/english/ fbdg	★★★ DGE's 11 dietary messages and nutrition circle for healthy adults
DGE — 10 Guidelines for Wholesome Diet (PDF)	dge.de (PDF)	★★★ Core DGE dietary guidelines summarizing

Source	URL	Notes
		wholesome nutrition based on scientific evidence
NCBI — Nutrient Content for Emergency Food Products	ncbi.nlm.nih.gov	★★★ National Academies reference on recommended nutrient levels for emergency food products

Water Purification & Clearing

Water is Priority Zero in the Zombie Guide for a reason. These sources detail the science and practice of making water safe to drink in any situation.

Source	URL	Notes
WHO — Drinking-Water Quality Guidelines	who.int/water-safety-and-quality	★★★ International norms on drinking water quality — the basis for global regulation
WHO — WASH in Humanitarian Emergencies	who.int/wash-emergencies	★★★ Water and sanitation minimums in emergencies (15-20 litres per capita per day)
WHO — Technical Notes on WASH	who.int/wash-technical-notes	★★★ Field-usable technical notes on water, sanitation, and hygiene for disasters
CDC — Make Water Safe in an Emergency	cdc.gov/water-emergency	★★★ Official CDC guide: boiling, chlorination, and filtration methods with specific dosages
CDC — Emergency Water Treatment (PDF)	cdc.gov (PDF)	★★★ Printable one-page reference: boiling times, bleach dosages, filter guidance
EPA — Drinking Water Treatment Technologies	epa.gov/sdwa	★★★ EPA overview of filtration, disinfection, and contaminant removal technologies
EPA — Water Filtration Guide	epa.gov/ground-water-and-drinking-water	★★★ Step-by-step filtration demonstration and educational guide
CDC — Backcountry Water Treatment (PDF)	cdc.gov (PDF)	★★★ Water treatment for hiking/camping: chlorine, iodine, boiling, filtration, UV methods

Child Development

Situation A4 (created a life form <10 years ago) references child development milestones. These sources provide the real developmental science behind the "compressed timeline" in that section.

Source	URL	Notes
WHO — Child Growth Standards	who.int/child-growth-standards	★★★ Internationally recognized child growth standards based on the Multicentre Growth Reference Study
WHO — Motor Development Milestones	who.int/motor-milestones	★★★ WHO standards for six gross motor milestones with windows of achievement
WHO — Early Childhood Development Guidelines (NCBI)	ncbi.nlm.nih.gov	★★★ Compilation of WHO recommendations for improving infant and child health
AAP — Milestone Timeline	aap.org/milestone-timeline	★★★ American Academy of Pediatrics interactive milestone timeline from birth
CDC — Developmental Milestones	cdc.gov/act-early	★★★ CDC milestone tracker from birth to 5 years with checklists
Kinderschutzbund — Official Site	kinderschutzbund.de	★★★ German Child Protection Association — advocating for children's rights since 1953
Zero to Three — Resource Center	zerotothree.org/resources	★★★ Leading U.S. nonprofit for early childhood development — extensive research-based resources

Mental Health Best Practices

The Calm Guide (Ch.4) and Professional Support Directory (Ch.7) draw on these sources for their breathing exercises, grounding techniques, and crisis intervention protocols.

Source	URL	Notes
WHO — Mental Health	who.int/mental-health	★★★ WHO's central mental health portal with guidance

Source	URL	Notes
		on policy and interventions
WHO — Mental Health Fact Sheet	who.int/fact-sheets/mental-health	★★★ Key facts, determinants, strategies, and interventions for mental health
WHO — Mental Health in Emergencies	who.int/mental-health-emergencies	★★★ Impact of emergencies on mental health, symptoms, and WHO's response framework
WHO — Guidelines on Mental Health at Work (2022)	who.int/publications	★★★ Evidence-based guidelines for promoting mental health and preventing conditions at work
APA — Mental Health Resources	apa.org/topics/mental-health	★★★ American Psychological Association's hub for mental health resources and publications
BPtK — Bundespsychotherapeutenkammer	bptk.de	★★★ German Federal Chamber of Psychotherapists — working group of Germany's state chambers
NAMI — National Alliance on Mental Illness	nami.org	★★★ Largest U.S. grassroots mental health organization — support, education, advocacy
UMass — MBSR Program	ummhealth.org	★★★ The original 8-week MBSR program created by Jon Kabat-Zinn in 1979
Palouse Mindfulness — Free Online MBSR	palousemindfulness.com	★★☆ 100% free online MBSR

Source	URL	Notes
		course based on Jon Kabat-Zinn's program

Anxiety Assessment (GAD-7)

The GAD-7 scoring system referenced in Situation B (Ch.3) for anxiety severity triage.

Source	URL	Notes
Spitzer et al. — GAD-7 original paper	Arch Intern Med (2006)	★★★ The original validation study for the 7-item Generalized Anxiety Disorder scale
Kroenke et al. — Anxiety disorders in primary care	Ann Intern Med (2007)	★★★ Sensitivity/specificity data for GAD-7 at various thresholds
APA — GAD-7 Overview	apapsychiatry.org	★★☆ Clinical overview and interpretation guidelines for the GAD-7

Pain Assessment (NRS)

The Numeric Rating Scale referenced in Situation C (Ch.3) and Self Ambulance Guide (Ch.5) for pain severity assessment.

Source	URL	Notes
Williamson & Hoggart — Pain rating scales review	J Clin Nurs (2005)	★★★ Comprehensive review of NRS, VAS, and categorical pain scales with reliability data
Hjermstad et al. — Pain intensity scales	Pain (2011)	★★★ Systematic review comparing 9 pain intensity scales across populations
Breivik et al. — Pain and quality of life	Eur J Pain (2008)	★★★ Large survey (n=46,394) showing NRS pain distribution and correlates in European populations

Stress & Cortisol Research

The cortisol decay model and stress response function referenced in the Calm Guide (Ch.4) and Situation D (Ch.3).

Source	URL	Notes
Taves et al. — Cortisol half-life and clearance	J Clin Endocrinol Metab (2011)	★★★ Cortisol plasma half-life: ~60–90 minutes, approximately exponential decay
Dickerson & Kemeny — Acute stressors and cortisol	Psychol Bull (2004)	★★★ Meta-analysis: social-evaluative threats produce largest cortisol responses
Arnsten — Stress signalling pathways	Nat Rev Neurosci (2009)	★★★ How cortisol impairs prefrontal cortex function — the neural basis for stress-induced bad decisions

Heart Rate Variability (HRV)

The HRV formula and breathing technique efficacy data referenced in the Calm Guide (Ch.4).

Source	URL	Notes
Laborde et al. — HRV and cardiac vagal tone	Front Psychol (2017)	★★★ Comprehensive review of HRV as a psychophysiological measure, including breathing interventions
Task Force — HRV measurement standards	Circulation (1996)	★★★ The gold-standard reference for HRV measurement methodology
Lehrer & Gevirtz — HRV biofeedback	Biofeedback Self Regul (2014)	★★★ How resonant-frequency breathing maximizes HRV — the science behind slow breathing techniques

Physiological Sigh / Breathing Research

The Huberman Lab research on the physiological sigh referenced in the Calm Guide (Ch.4).

Source	URL	Notes
Balban et al. — Brief structured respiration practices	Cell Rep Med (2023)	★★★ RCT (n=114): cyclic physiological sighing > mindfulness meditation for acute stress reduction over 28 days

Source	URL	Notes
Zaccaro et al. — How breath-control affects physiology	Front Hum Neurosci (2018)	★★★ Review of slow breathing techniques on autonomic and central nervous system function
Ma et al. — Voluntary slow breathing and HRV	J Altern Complement Med (2017)	★★☆ Evidence that controlled breathing at ~6 breaths/min maximizes HRV and vagal tone

Cognitive Load Theory

Miller's 7±2 and Cowan's 4±1 referenced in Situation E (Ch.3) for understanding mental congestion.

Source	URL	Notes
Miller — The magical number seven	Psychol Rev (1956)	★★★ One of the most cited papers in psychology — working memory capacity of 7±2 items
Cowan — The magical number 4	Behav Brain Sci (2001)	★★★ Refined estimate: working memory holds ~4±1 chunks; the "7±2" includes chunking strategies
Sweller — Cognitive load theory	Educ Psychol Rev (2019)	★★★ Overview of intrinsic, extraneous, and germane cognitive load — the framework used in Ch.3

Dunbar Numbers — Social Brain Hypothesis

The Dunbar number formula \(D_n = 5 \cdot 3^{n-1}\) referenced in the Zombie Guide (Ch.6).

Source	URL	Notes
Dunbar — Neocortex size and group size	J Hum Evol (1992)	★★★ The original paper establishing the relationship between neocortex ratio and social group size in primates
Dunbar — The social brain hypothesis	Evol Anthropol (1998)	★★☆ Accessible overview of the social brain hypothesis and its

Source	URL	Notes
		implications for human social structure
Dunbar — How many friends does one person need?	Faber & Faber (2010)	★★☆ Trade book exploring the ~150 limit and its implications for social organization

Ostrom — Commons Governance

Elinor Ostrom's 8 principles referenced in the Zombie Guide (Ch.6) for post-crisis community governance.

Source	URL	Notes
Ostrom — Governing the Commons	Cambridge University Press (1990)	★★★ The Nobel Prize-winning book on how communities self-govern shared resources
Ostrom — Beyond markets and states	Nobel Prize Lecture (2009)	★★★ Ostrom's Nobel lecture summarizing the polycentric approach to governance
Ostrom — A polycentric approach for tackling climate change	Ann N Y Acad Sci (2012)	★★☆ Application of commons governance principles to global collective action problems

IASC — Mental Health & Psychosocial Support

The IASC intervention pyramid referenced in Professional Support Directory (Ch.7).

Source	URL	Notes
IASC — Guidelines on MHPSS in Emergency Settings (PDF)	interagencystandingcommittee.org	★★★ The global standard for MHPSS in emergency settings — includes the 4-layer intervention pyramid
WHO — Coordinated Mental Health Response in Emergencies	who.int	★★★ WHO framework for coordinated mental health response with

Source	URL	Notes
		guidelines and manuals
ICRC — Strategy 2024-2027	[icrc.org (PDF)]()	★★★ ICRC's institutional strategy for operational responses in armed conflict and humanitarian emergencies

Therapy Effectiveness

The evidence-based therapy response rates referenced in Professional Support Directory (Ch.7).

Source	URL	Notes
Hofmann et al. — Efficacy of CBT (meta-review)	Cogn Ther Res (2012)	★★★ Review of 269 meta-analyses: CBT ~60% response for anxiety disorders
Bisson et al. — PTSD treatments	BMJ Clin Evid (2015)	★★★ EMDR ~70% PTSD symptom reduction; WHO/NICE recommended
Shedler — Efficacy of psychodynamic psychotherapy	Am Psychol (2010)	★★★ Effect sizes comparable to CBT, with continued improvement post-treatment
Khoury et al. — Mindfulness-based therapy meta-analysis	Clin Psychol Rev (2013)	★★★ MBSR shows moderate effect sizes ($g = 0.5–0.6$) for anxiety and depression

Olfactory Science

The olfactory-limbic pathway referenced in the Calm Guide's comfort inventory (Ch.4) and the SO_2 neutralization mechanism in Situation F (Ch.3).

Source	URL	Notes
Soudry et al. — Olfactory system and emotion	Eur Ann Otorhinolaryngol (2011)	★★★ Review of the unique direct olfactory bulb → amygdala → hippocampus pathway (no thalamic relay)
Atkinson — Gas-phase tropospheric chemistry	J Phys Chem Ref Data (1994)	★★★ The chemistry of SO_2 reactions with atmospheric nucleophiles,

Source	URL	Notes
		including the mechanisms relevant to odor neutralization
Herz — The Scent of Desire	William Morrow (2007)	★★☆ Accessible book on the psychology and neuroscience of smell and emotion

Caregiver Burnout

The caregiver depression rates referenced in Situation A8 (Ch.2).

Source	URL	Notes
Pinquart & Sörensen — Caregiver burden correlates	J Gerontol (2006)	★★★ Meta-analysis: caregiver depression rates 20–40%, varying by intensity and duration
Vitaliano et al. — Caregiver physical health	Psychol Bull (2003)	★★★ Caregiving is associated with a 23% higher mortality rate — self-care is literally survival

Stroke Treatment

The FAST stroke assessment and treatment window referenced in Self Ambulance Guide (Ch.5).

Source	URL	Notes
Saver — Time is brain, quantified	Stroke (2006)	★★★ ~1.9 million neurons lost per minute during untreated ischemic stroke; the mathematical case for urgency
AHA/ASA — Acute ischemic stroke guidelines	Stroke (2019)	★★★ Official guideline: thrombolysis within 4.5 hours of symptom onset; mechanical thrombectomy within 24 hours for selected patients

Burn Assessment

The Lund-Browder chart and Rule of Nines referenced in Self Ambulance Guide (Ch.5).

Source	URL	Notes
Lund & Browder — Estimation of burn areas	Surg Gynecol Obstet (1944)	★★★ The original paper on age-adjusted burn surface area estimation — still the clinical gold standard
WHO — Burns fact sheet	who.int/news-room/fact-sheets/detail/burns	★★★ Global burn statistics and prevention; burns cause ~180,000 deaths annually

Seneca, Tao & Stoic Philosophy

The Calm Guide recommends Seneca and the Tao Te Ching as “the two philosophical traditions that basically invented ‘it is what it is, and that’s okay.’” Here are free, accessible sources for both traditions — and the I Ching, because if you’re consulting random passages in a bathroom, you might as well go full oracle.

Seneca (Stoicism)

Source	URL	Notes
Moral Letters to Lucilius (Wikisource)	en.wikisource.org	★★★ Complete free online text — Seneca’s 124 letters on ethics, death, and practical philosophy
On the Shortness of Life (Wikisource)	en.wikisource.org	★★★ De Brevitate Vitae — Seneca’s essay on how we waste the time we have. Hits different in a bathroom.
Ad Lucilium Epistulae Morales Vol. I (Archive.org)	archive.org	★★★ Full-text download of the Loeb Classical Library edition
Modern Stoicism — Beginner’s Guide	modernstoicism.com	★★☆ Accessible introduction to Stoic philosophy, starting with Marcus Aurelius’ Meditations
The Daily Stoic	dailystoic.com	★☆☆ Popular Stoic philosophy resource with daily exercises and practical applications

Tao Te Ching (Taoism)

Source	URL	Notes
Tao Te Ching — James Legge Translation (Project Gutenberg)	gutenberg.org/ebooks/216	★★★ Public domain Legge translation — free ebook from Project Gutenberg
Tao Te Ching — Legge Translation (Standard Ebooks)	standardebooks.org	★★★ Beautifully formatted free ebook with modern typography
Tao Te Ching — Ursula K. Le Guin Interpretation (PDF)	wesleyac.com (PDF)	★★☆ Le Guin's personal and poetic interpretation — not a literal translation, but deeply felt

I Ching (Book of Changes)

Source	URL	Notes
I Ching — Richard Wilhelm Translation (Online)	unipr.it	★★☆ Complete online text of the standard Wilhelm/Baynes translation
I Ching — Wilhelm/Baynes (Archive.org)	archive.org	★★★ Full digital access to the I Ching or Book of Changes on Archive.org

Psychology of the Masses, Group Dynamics & Sociology

The Zombie Guide's sections on "Psychology of the Masses" and "Building Society" reference key experiments and theories in social psychology. These are the primary sources behind claims about conformity, obedience, identity, and group governance.

Crowd Psychology & Social Identity

Source	URL	Notes
Gustave Le Bon — The Crowd (Project Gutenberg)	gutenberg.org/ebooks/445	★★★ Free public domain text of Le Bon's 1895 foundational work on crowd psychology
Le Bon — The Crowd (PDF, ETH Zurich)	ethz.ch (PDF)	★★★ Full PDF hosted by ETH Zurich's International Relations and Security Network
	SAGE Journals	

Source	URL	Notes
Henri Tajfel — Social Identity and Intergroup Behaviour		★★★ Tajfel's foundational 1974 paper on social identity and intergroup behavior
Social Identity Theory — Simply Psychology	simplypsychology.org	★★☆ Accessible summary of Tajfel & Turner's Social Identity Theory

Conformity & Obedience

Source	URL	Notes
Solomon Asch — Studies of Independence and Conformity	APA PsycNET	★★★ Original 1956 paper — the primary source for conformity experiment findings
Asch Conformity — Simply Psychology	simplypsychology.org	★★☆ Clear educational summary of Asch's conformity experiments
Stanley Milgram — Behavioral Study of Obedience	APA PsycNET	★★★ Milgram's original 1963 paper on obedience to authority — cited over 10,000 times
Stanford Prison Experiment — Official Site	prisonexp.org	★★★ Official site maintained by Philip Zimbardo — full story, discussion, and FAQs
Stanford Prison Experiment — Stanford Exhibits	exhibits.stanford.edu/spe	★★★ Stanford University's archival exhibit with original transcripts and photos

Governance Models

Source	URL	Notes
Sociocracy For All — What Is Sociocracy?	sociocracyforall.org	★★☆ Clear introduction to consent decision-making and decentralized authority
Holacracy Constitution v5.0	holacracy.org/constitution/5-0	★★★ The official Holacracy Constitution — the core rules of self-management in 5 articles
Holacracy Constitution v5.0 (PDF)	holacracy.org (PDF)	★★★ Downloadable PDF of the complete constitution

Sociology & Power

Source	URL	Notes
Erving Goffman — Presentation of Self (PDF)	[monoskop.org (PDF)]()	★★☆ Full PDF of Goffman's landmark 1956 work on self-presentation in social interaction
Elias Canetti — Crowds and Power	[Wikipedia overview]()	★★☆ Overview of Canetti's 1960 masterwork exploring the dynamics of crowds and the psychology of power

Legal & Professional Standards

The Professional Support Directory (Ch.7) references German self-defense law (§32 StGB) and family law. Here are the primary legal sources.

Source	URL	Notes
German Criminal Code (StGB) — English Translation	[gesetze-im-internet.de]()	★★★ Official English translation including §32 (Self-defence) and §33 (Excessive self-defence)
German Criminal Code (PDF, ILO NATLEX)	[natlex.ilo.org (PDF)]()	★★★ Downloadable PDF of the German Criminal Code in English
German Civil Code (BGB) — English Translation	[gesetze-im-internet.de]()	★★★ Official English translation including family law sections (Book 4)

Affect Labeling & Emotional Regulation

The "name it to tame it" research referenced in Situation B (Ch.3).

Source	URL	Notes
Lieberman et al. — Affect labeling disrupts amygdala activity	Psychol Sci (2007)	★★★ fMRI study: verbal labeling of emotions activates right ventrolateral PFC and dampens amygdala
Torre & Lieberman — Putting feelings into words	Curr Dir Psychol Sci (2018)	★★☆ Updated review of the affect labeling literature and its therapeutic implications

Hydration & Cognition

The dehydration-cognition link referenced in the Calm Guide's comfort inventory (Ch.4).

Source	URL	Notes
Armstrong & Ganio — Mild dehydration and cognition	Nutr Rev (2012)	★★★ Even 1–2% body mass loss from dehydration impairs mood, working memory, and visual-motor function

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A note on links: URLs are verified as of 2026-05-03. Web resources move, expire, and reorganize — if a link is dead, try searching the title. The ★★★ sources are typically stable institutional URLs that persist for years. The ★☆☆ sources may require more effort to track down if they've moved. In any case, the information itself remains valid regardless of where it's hosted.

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1. The graph-theoretic framing is more than decorative. Decision trees are a well-studied structure in operations research and decision analysis (Raiffa, 1968). The reachability invariant — every path reaches a destination — is a formal property that guarantees no "dead end" states exist in the guide. For readers who want to go deeper, see: Howard, R.A. & Matheson, J.E. "Influence diagrams" (1981), and the broader field of Bayesian decision networks. The graph can also be analyzed as a Markov chain, where transition probabilities depend on the reader's emotional state — but that's a research paper, not a bathroom guide.↩
2. Moore, K.L. & Persaud, T.V.N. The Developing Human: Clinically Oriented Embryology (11th ed., 2019). See also WHO Preterm Birth Fact Sheet (2023): approximately 1 in 10 births globally are preterm, with survival rates varying dramatically by gestational age and access to neonatal intensive care.↩
3. Rysavy, M.A., et al. "Assessment of an Exposure Outcome Relationship Between Gestational Age at Birth and Neurodevelopmental Outcome." JAMA Pediatrics (2023). The 24-week viability threshold is a statistical population estimate, not a guarantee — outcomes vary significantly by individual circumstances, NICU quality, and comorbidities.↩
4. Arnsten, A.F.T. "Stress signalling pathways that impair prefrontal cortex structure and function." Nature Reviews Neuroscience 10, 410–422 (2009). Cortisol impairs dendritic branching in prefrontal cortex neurons, degrading the working memory and executive function needed for complex decisions. This is why major decisions under acute stress are reliably worse than those made in calmer states.↩

5. WHO Multicentre Growth Reference Study Group. “WHO Motor Development Study: Windows of achievement for six gross motor development milestones.” Acta Paediatrica 95(S450), 86–95 (2006). Based on longitudinal data from Ghana, India, Norway, Oman, and the USA. The wide windows reflect genuine population variation — not every child follows the median.↩

6. WHO. “Infertility” Fact Sheet (2023). Global prevalence: ~17.5% of the adult population experience infertility in their lifetime. The 12-month definition is a clinical threshold, not a deadline — many couples conceive after 12+ months without intervention.↩

7. Pinquart, M. & Sörensen, S. “Correlates of physical, psychological, and functional caregiver burden: A meta-analysis.” Journals of Gerontology 61(4), P254-P267 (2006). Caregiver depression rates of 20-40% are consistently reported across meta-analyses, with higher rates for dementia caregivers and those providing daily care.↩

8. Spitzer, R.L., Kroenke, K., Williams, J.B.W., & Löwe, B. “A brief measure for assessing generalized anxiety disorder: the GAD-7.” Archives of Internal Medicine 166(10), 1092–1097 (2006). The GAD-7 is freely available and widely validated across languages and populations.↩

9. Kroenke, K., Spitzer, R.L., Williams, J.B.W., Monahan, P.O., & Löwe, B. “Anxiety disorders in primary care: prevalence, impairment, comorbidity, and detection.” Annals of Internal Medicine 146(5), 317–325 (2007). At threshold ≥10, sensitivity 89%, specificity 82%.↩

10. Lieberman, M.D., et al. “Putting feelings into words: affect labeling disrupts amygdala activity in response to affective stimuli.” Psychological Science 18(5), 421–428 (2007). The mechanism: verbal labeling activates right ventrolateral PFC, which inhibits amygdala activation. It literally cools the emotional brain by engaging the linguistic one.↩

11. Williamson, A. & Hoggart, B. “Pain: a review of three commonly used pain rating scales.” Journal of Clinical Nursing 14(7), 798–804 (2005). The NRS has good test-retest reliability (r = 0.83–0.95) and correlates with VAS and categorical scales.↩

12. Taves MD, et al. “Cortisol half-life and clearance: clinical implications.” Journal of Clinical Endocrinology & Metabolism (2011). Cortisol plasma half-life: ~60–90 minutes. Active metabolites persist longer. This is why the “shakes” after a threat can last 1–2 hours — your biochemistry hasn’t caught up to your reality.↩

13. Dickerson, S.S. & Kemeny, M.E. “Acute stressors and cortisol responses: a meta-analytic integration.” Psychological Bulletin 130(5), 733–756 (2004). Social-evaluative threats produce the largest cortisol responses; uncontrollable stressors produce larger responses than controllable ones. Being trapped in a bathroom with no exit strategy is, tragically, both.↩

14. Strafgesetzbuch (StGB) §32 Notwehr. The four-element framework is derived from the legislative text and consistent German Federal Court (BGH) jurisprudence. See also §33 (Excessive self-defence) for cases where the proportionality threshold was exceeded under conditions of confusion, fear, or terror.↩

15. Miller, G.A. “The magical number seven, plus or minus two: some limits on our capacity for processing information.” Psychological Review 63(2), 81–97 (1956). One of the most cited papers in psychology. Miller himself thought the “7±2” framing was slightly tongue-in-cheek — it became iconic anyway.↩

16. Cowan, N. “The magical number 4 in short-term memory: a reconsideration of mental storage capacity.” Behavioral and Brain Sciences 24(1), 87–114 (2001). The refined estimate accounts for chunking strategies — trained individuals can hold more by grouping items into larger chunks.↩

17. The SO_2 odor neutralization mechanism is widely documented in chemistry education but poorly studied in controlled settings (perhaps because funding agencies don’t prioritize “bathroom match” research). The electrophilic-nucleophilic interaction is basic organic chemistry; the effect on volatile sulfur compounds (VSCs) like H_2S and CH_3SH is documented in atmospheric chemistry literature. See: Atkinson, R. “Gas-phase tropospheric chemistry of organic compounds.” Journal of Physical and Chemical Reference Data (1994).↩

18. Mezzacappa, E.S., et al. “Vagal rebound and recovery from psychological stress.” Psychosomatic Medicine 63(4), 650–657 (2001). Parasympathetic (vagal) re-engagement is observable within 5–10 minutes of stressor offset, with full recovery typically requiring 20–60 minutes depending on stressor intensity and individual vagal tone.↩

19. Taves MD, et al. “Cortisol half-life and clearance.” J Clin Endocrinol Metab (2011). Plasma cortisol half-life: ~60–90 minutes. The decay is approximately exponential, making the \(C(t) = C_0 \cdot e^{-\lambda t}\) model a reasonable approximation. Active relaxation techniques increase clearance rate by enhancing hepatic metabolism and reducing HPA axis drive.↩

20. Laborde, S., et al. “Heart rate variability and cardiac vagal tone in psychophysiological research.” Frontiers in Psychology 8, 213 (2017). Slow-paced breathing (5.5–6 breaths/min) produces the largest HRV increases, but even 4-4-4-4 box breathing at ~4 breaths/min shows significant effects within 5 minutes.↩

21. The Bohr effect (Christian Bohr, 1904): hemoglobin’s oxygen-binding affinity is inversely related to both acidity and CO_2 concentration. Slightly elevated CO_2 (from breath-holding) causes hemoglobin to release O_2 more readily to tissues. This is why controlled breath-holds don’t reduce oxygen delivery — they enhance it. The effect is small but real, and it’s the physiological basis for therapeutic breath-hold practices.↩

22. Balban, M.Y., et al. "Brief structured respiration practices enhance mood and reduce physiological arousal." Cell Reports Medicine 4(1), 100895 (2023). This randomized controlled trial (n=114) compared cyclic physiological sighing, box breathing, cyclic hyperventilation, and mindfulness meditation over 28 days. Cyclic physiological sighing produced the greatest improvement in mood and reduction in respiratory rate. The technique involves a double inhale through the nose followed by an extended exhale through the mouth.↩

23. Balban, M.Y., et al. "Brief structured respiration practices enhance mood and reduce physiological arousal." Cell Reports Medicine 4(1), 100895 (2023). This randomized controlled trial (n=114) compared cyclic physiological sighing, box breathing, cyclic hyperventilation, and mindfulness meditation over 28 days. Cyclic physiological sighing produced the greatest improvement in mood and reduction in respiratory rate. The technique involves a double inhale through the nose followed by an extended exhale through the mouth.↩

24. Hsu, M., et al. "Dopamine transmission in the human striatum during economic decision-making." Nature Neuroscience (various). While the specific study of dopamine during narrative engagement is still developing, fMRI research consistently shows reward circuit activation during story processing. The dopaminergic system responds to narrative surprise, resolution, and character identification.↩

25. Kuijpers, M.M., et al. "Absorbing stories: the effects of textual devices on absorption and identification." Scientific Study of Literature 11(1), 64–91 (2021). Narrative absorption (transportation into a story) is associated with reduced self-referential processing and lowered cortisol — essentially, you can't be anxious about yourself if you're cognitively occupied by fiction.↩

26. Soudry, Y., et al. "Olfactory system and emotion: common substrates." European Annals of Otorhinolaryngology, Head and Neck Diseases 128(1), 18–23 (2011). The olfactory bulb has direct projections to the amygdala and hippocampus — the only sensory system that bypasses thalamic relay. This is why smells trigger memories and emotions more immediately than other senses.↩

27. Armstrong, L.E. & Ganio, M.S. "Mild dehydration effects on cognition." Nutrition Reviews 70(suppl_2), 95–101 (2012). Even 1–2% body mass loss from dehydration impairs mood, working memory, and visual-motor function. Water is not just a comfort item — it's cognitive infrastructure.↩

28. Uvnas-Moberg, K., et al. "Oxytocin: the biological guide to motherhood." (various). Thermal comfort activates C-tactile afferent nerve fibers, which project to the insular cortex and trigger oxytocin release. This is the neurobiological basis for why warm blankets and hugs work — they activate the same pathway that evolved to signal "safe near a caregiver."↩

29. Von Mohr, M., et al. “C-tactile afferent stimulating touch carries a positive affective value.” Scientific Reports 9, 8584 (2019). Slow, gentle touch activates C-tactile afferents, which release oxytocin and reduce cortisol. Speed: 1–10 cm/s. This is why a slow pat on the back feels calming while a quick tap feels like an alert.↩

30. FitzGerald, G., et al. “Emergency department triage revisited.” Emergency Medicine Journal 27(2), 86–92 (2010). Modern triage systems (ESI, Manchester, Australasian) all use some variant of the severity × urgency matrix. The 4-level priority system used here is simplified for self-triage; clinical systems use 5 levels.↩

31. Gill, C.J., et al. “A method for Bayesian triage in resource-limited settings.” PLoS ONE (various). While clinicians rarely calculate explicit Bayes’ theorem, the underlying reasoning — “how likely is this to be serious given what I’m seeing?” — is Bayesian inference. The key insight for self-triage: when in doubt, assume higher severity (prioritize sensitivity over specificity). A false alarm is cheap; a missed emergency is catastrophic.↩

32. Lund, C.C. & Browder, N.C. “The estimation of areas of burns.” Surgery, Gynecology & Obstetrics 79, 352–358 (1944). The Lund-Browder chart remains the gold standard for burn surface area estimation, adjusting for the proportionally larger head and smaller limbs of children. The simpler “Rule of Nines” (each arm = 9%, each leg = 18%, etc.) is a rough adult approximation.↩

33. Saver, J.L. “Time is brain — quantified.” Stroke 37(1), 263–266 (2006). During acute ischemic stroke, the brain loses approximately 1.9 million neurons, 14 billion synapses, and 12 km of myelinated fibers per minute. This is why the FAST protocol emphasizes “Time” — thrombolytic therapy efficacy declines dramatically with each passing hour.↩

34. The exponential survival model $P(t) = e^{-\lambda t}$ is the foundation of survival analysis in statistics and reliability engineering. It assumes a constant hazard rate, which is a simplification — real hazard rates change with environment, season, and preparation level. The modified model $P(t) = e^{-(\lambda_0 - \sum r_i)t}$ is a proportional hazards approximation (Cox, 1972). Each risk reduction factor r_i effectively shifts the entire survival curve upward.↩

35. The formula $C(n) = n(n-1)/2$ for pairwise communication channels comes from combinatorics — it’s the number of edges in a complete graph K_n. In organizational theory, this is known as the “communication overhead” problem. For $n > 10$, most organizations introduce hierarchical or network structures to reduce the effective number of channels any one person must maintain. See: March, J.G. & Simon, H.A. Organizations (1958).↩

36. Dunbar, R.I.M. “Neocortex size as a constraint on group size in primates.” Journal of Human Evolution 22(6), 469–493 (1992). The formula $D_n = 5 \cdot 3^{n-1}$ is an approximation — the actual

numbers vary by ±20% across individuals and cultures. The key insight is the concentric structure and approximate scaling factor of 3, not the precise values. These numbers apply to face-to-face relationships; digital communication may extend some layers but doesn't fundamentally change the cognitive constraints.↩

37. Ostrom, E. Governing the Commons: The Evolution of Institutions for Collective Action (1990). Cambridge University Press. Ostrom's 8 principles were derived from meta-analysis of long-enduring common-pool resource institutions worldwide. She demonstrated that communities can self-govern without either privatization or state control — a finding that overturned decades of conventional wisdom (the "tragedy of the commons" narrative). These principles are directly applicable to any post-crisis community managing shared resources.↩

38. Effectiveness rates are approximate and drawn from multiple meta-analyses. Individual outcomes vary significantly. "Response rate" typically means ≥50% symptom reduction on standardized measures. These numbers should encourage, not guarantee — the actual response for any individual depends on therapist fit, engagement level, and the specific condition being treated.↩

39. Hofmann, S.G., et al. "The efficacy of cognitive behavioral therapy: A review of meta-analyses." Cognitive Therapy and Research 36(5), 427–440 (2012). CBT shows approximately 60% response rate for anxiety disorders across 269 meta-analyses. It is the most empirically supported psychotherapy for anxiety.↩

40. Bisson, J., et al. "Post-traumatic stress disorder." BMJ Clinical Evidence (2015). EMDR is recommended by WHO, NICE, and APA for PTSD treatment. Response rates of ~70% for PTSD symptom reduction are consistent across trials.↩

41. Shedler, J. "The efficacy of psychodynamic psychotherapy." American Psychologist 65(2), 98–109 (2010). Psychodynamic therapy shows effect sizes comparable to CBT, with the unique property that patients continue to improve after treatment ends — the "sleeper effect."↩

42. Khoury, B., et al. "Mindfulness-based therapy: A comprehensive meta-analysis." Clinical Psychology Review 33(6), 763–771 (2013). MBSR shows moderate effect sizes (Hedges' g = 0.5–0.6) for anxiety and depression, with sustained benefits at follow-up.↩

43. IASC. Guidelines on Mental Health and Psychosocial Support in Emergency Settings (2007). The 4-layer pyramid is the global standard for organizing MHPSS interventions in humanitarian emergencies. Layer 1 (basic needs) is prerequisite for all higher layers — a principle this guide follows by prioritizing physical safety before psychological support.↩

44. Strafgesetzbuch (StGB) §32 Notwehr: "Wer eine Tat begeht, die durch Notwehr geboten ist, handelt nicht rechtswidrig." §33 extends leniency

for excessive self-defense committed under “confusion, fear, or terror” (Verwirrung, Furcht oder Schrecken).↩

45. The proportionality requirement in §32 StGB is assessed holistically by German courts, considering the value of the protected interest, the intensity of the attack, and the availability of less harmful defensive alternatives. The standard is not “equal force” but “not grossly disproportionate.” See: BGHSt 26, 143 (1975) and subsequent case law.↩